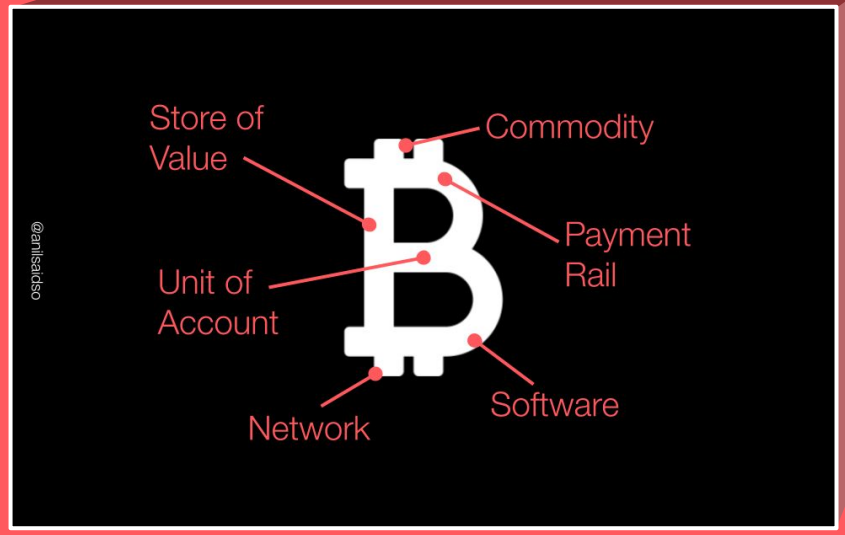
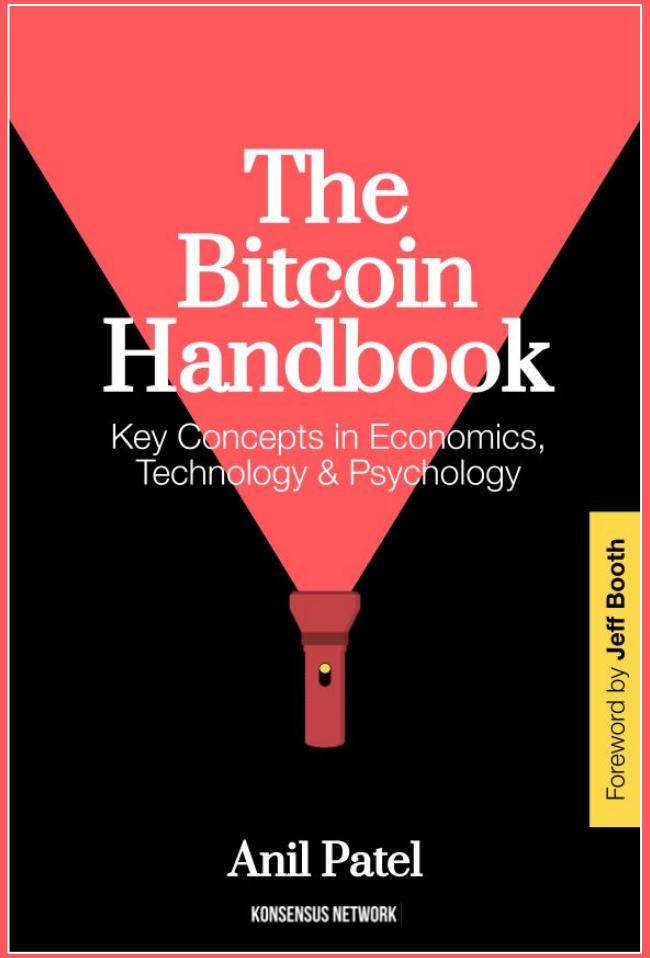
The Bitcoin Handbook

PRESENTATION DECK



Anil Patel

This deck is adapted from the book:





Anil

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Contents

I. Economics

1. Scarcity
2. Monetary Premium
3. Gresham's Law
4. Thier's Law
5. Cantillon Effect
6. Schelling Point
7. Opportunity Cost
8. Time Preference
9. Impossible Trinity
10. Jevons Paradox
11. Power Laws
12. Winner-Take-All Effects
13. Unit Bias
14. Veblen Good
15. Malinvestment
16. Asymmetric Payoff
17. Ansoff Matrix

II. Technology & Systems

1. Gartner Hype Cycle
2. Amara's Law
3. Orders of Magnitude
4. 10x Improvement Rule
5. Network Effects
6. Churn
7. Accelerating Returns
8. Higher-Order Effects
9. Creative Destruction
10. Feedback Loops
11. Reflexivity
12. Relativity
13. Thermodynamics 1st law
14. Thermodynamics 2nd law
15. Information Theory
16. Arrow of Time
17. Moore's Law
18. Antifragility
19. Gall's Law
20. Attack Surface
21. Catalysts
22. Friction

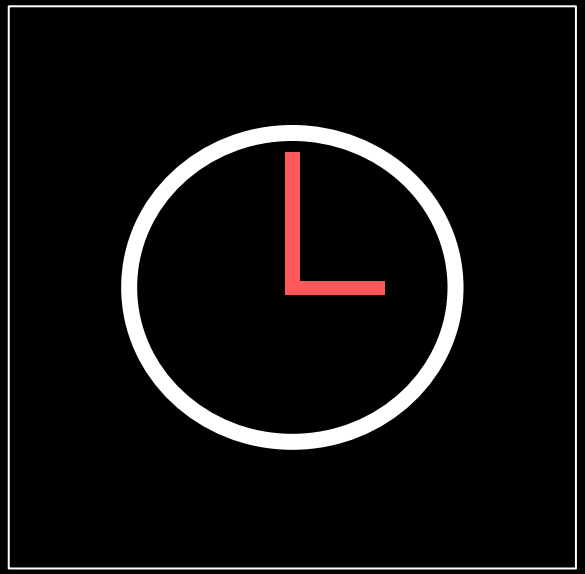
III. Psychology

1. Prisoner's Dilemma
2. Jurisdictional Arbitrage
3. Social Proof
4. Mimetic Theory
5. First Conclusion Bias
6. Circle of Competence
7. Dunning-Kruger Effect
8. Category Error
9. Intransigent Minority
10. Inequivalence Theorem
11. Gell-Mann Amnesia
12. Lindy Effect
13. Emergence
14. Streisand Effect
15. Moral Hazard
16. Counterparty Risk

Economics

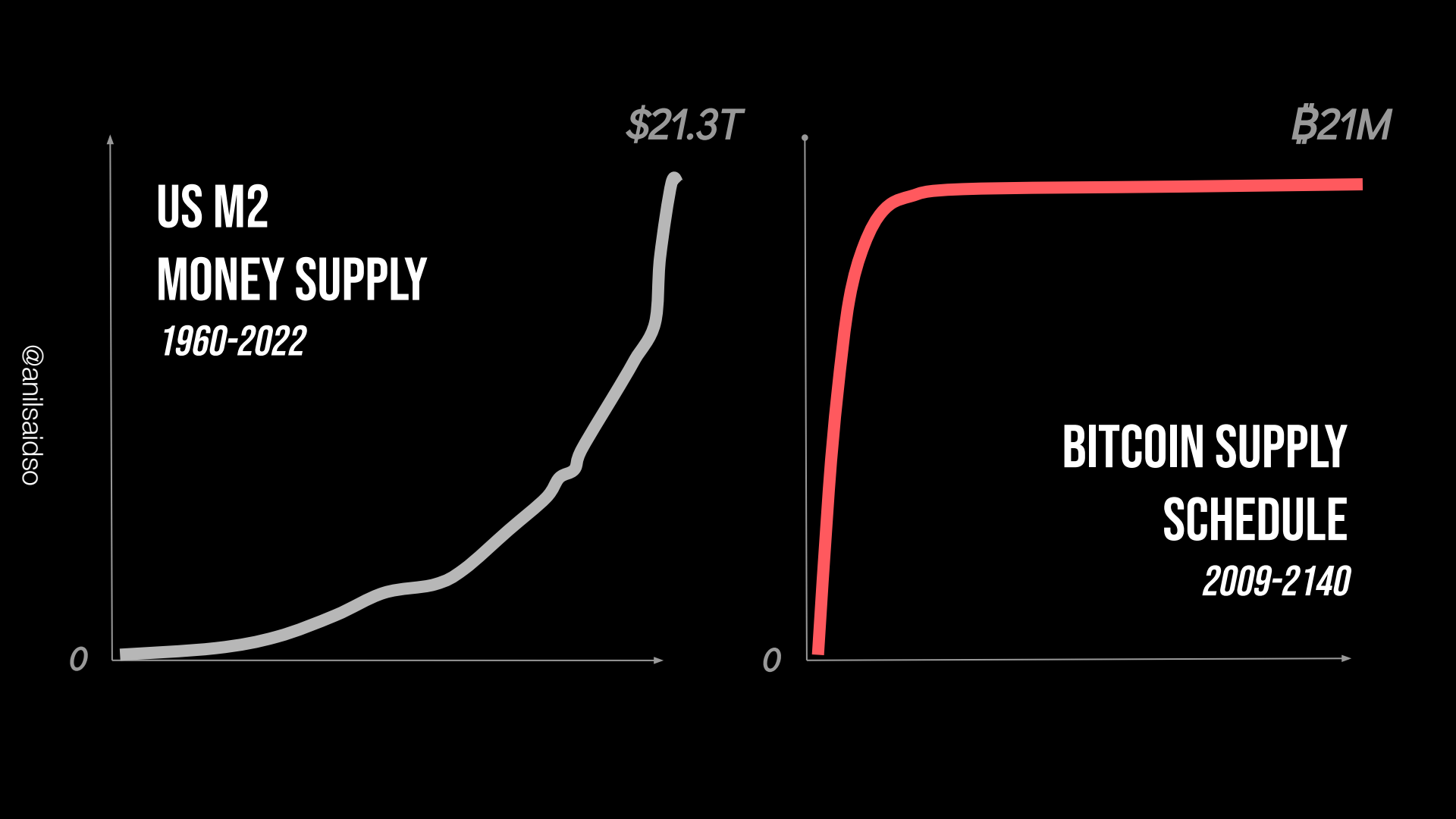
SCARCITY

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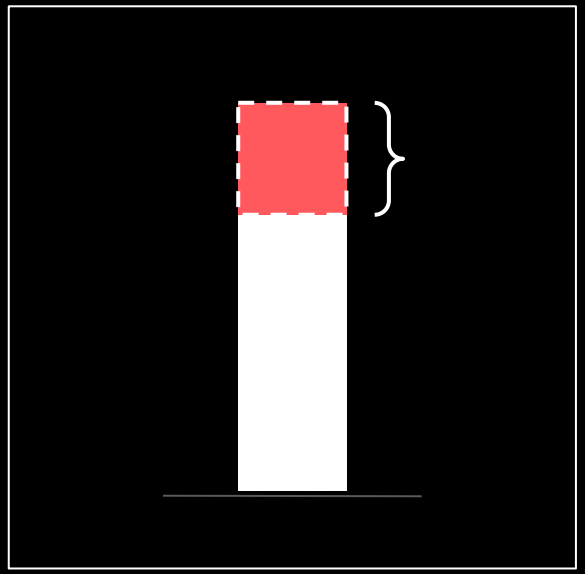
How limited resources are allocated and competed for

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$21.3T
US M2
MONEY SUPPLY
1960-2022
0
₿21M
BITCOIN SUPPLY
SCHEDULE
2009-2140
0
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MONETARY PREMIUM

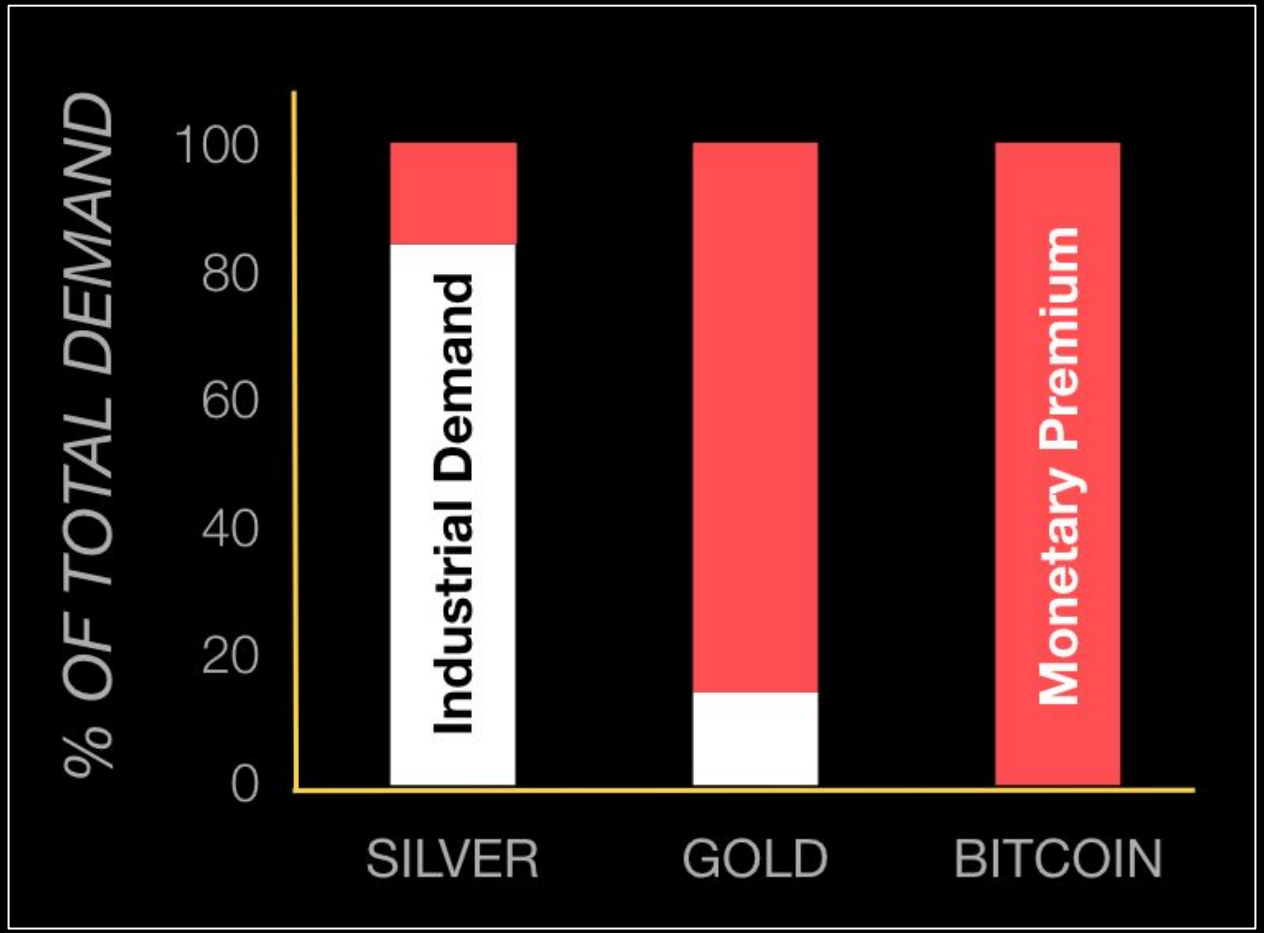
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Purchasing power of a monetary good in excess of its use-value alone

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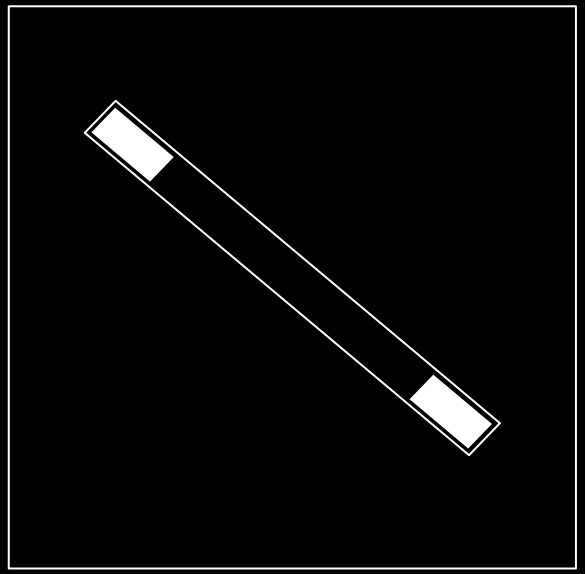
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source: *'The Bullish Case for Bitcoin'* by Vijay Boyapati

GRESHAM'S LAW

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When two forms of commodity money are decreed with equivalent face value, the more valuable commodity will be saved—disappearing from circulation.



More valuable ⟷ *Less valuable*

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THIER'S LAW

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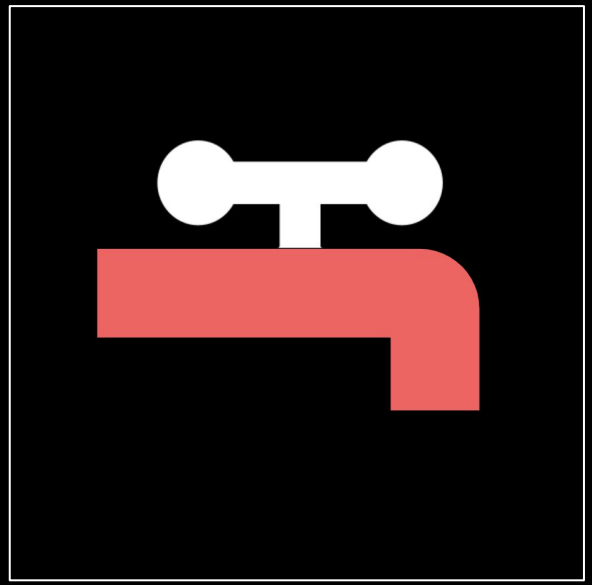


Absent legal tender laws, money that cannot be relied upon to preserve value will be rejected in favor of superior forms.

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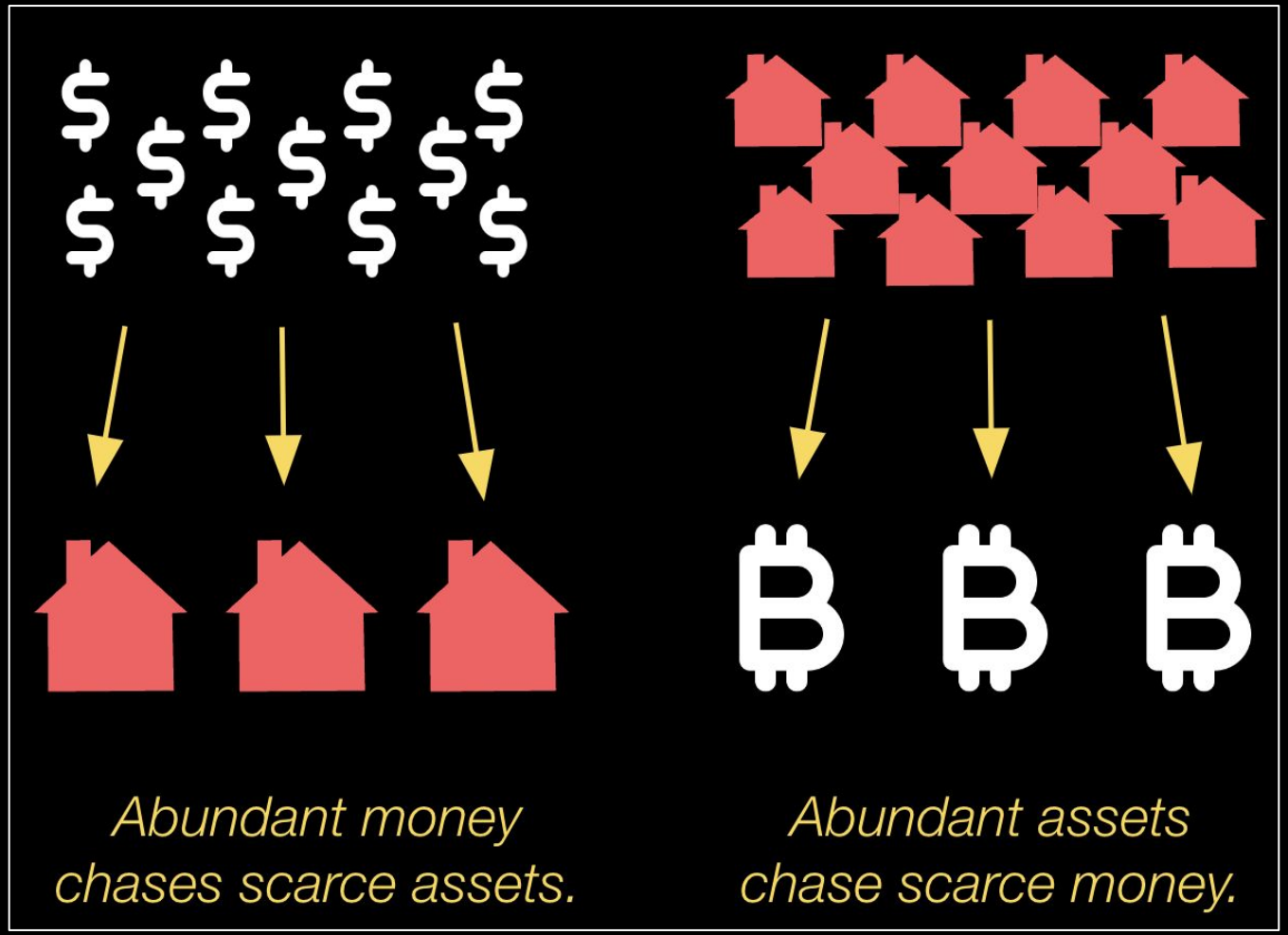
CANTILLON EFFECT

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New units of money create disproportionate price inflation when added into an economy based on the path they travel

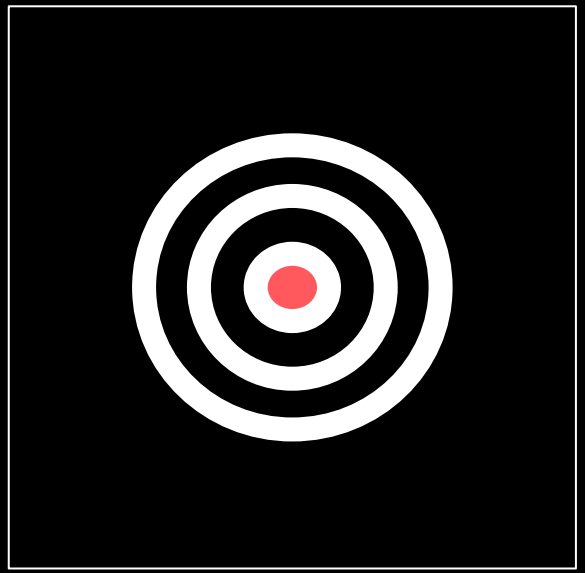
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Abundant money
chases scarce assets.
Abundant assets
chase scarce money.

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SCHELLING POINT

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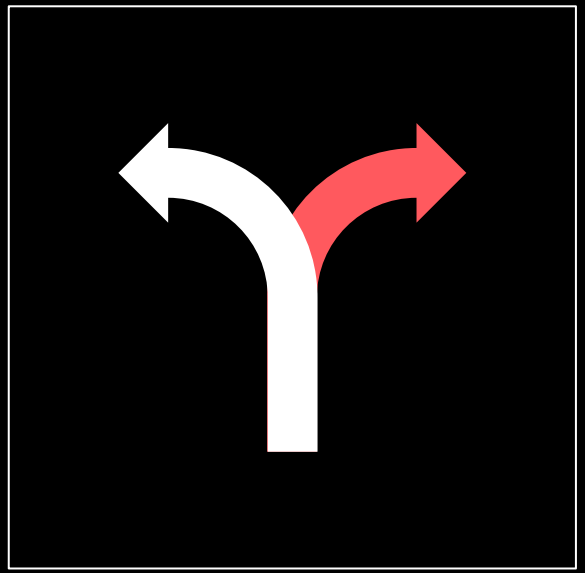


A solution that people tend to choose, by default, in the absence of communication

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OPPORTUNITY COST

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When doing one thing comes at the cost of not being able to do something else

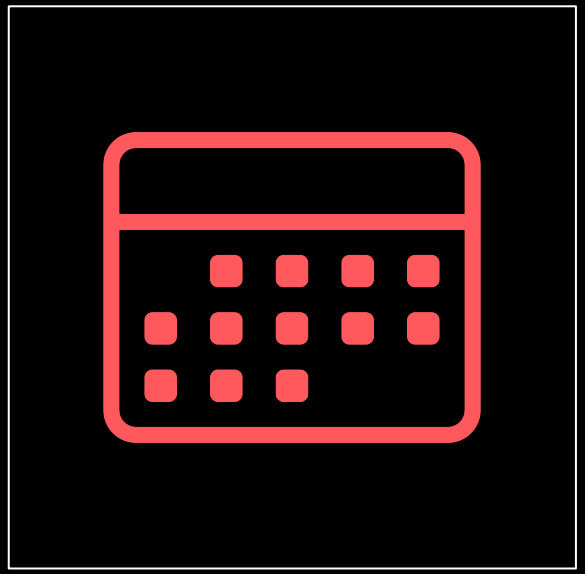
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“When money is hard and can appreciate, individuals are likely to be very discerning about what they spend it on, as the opportunity cost appreciates over time.”

—SAIFEDEAN AMMOUS

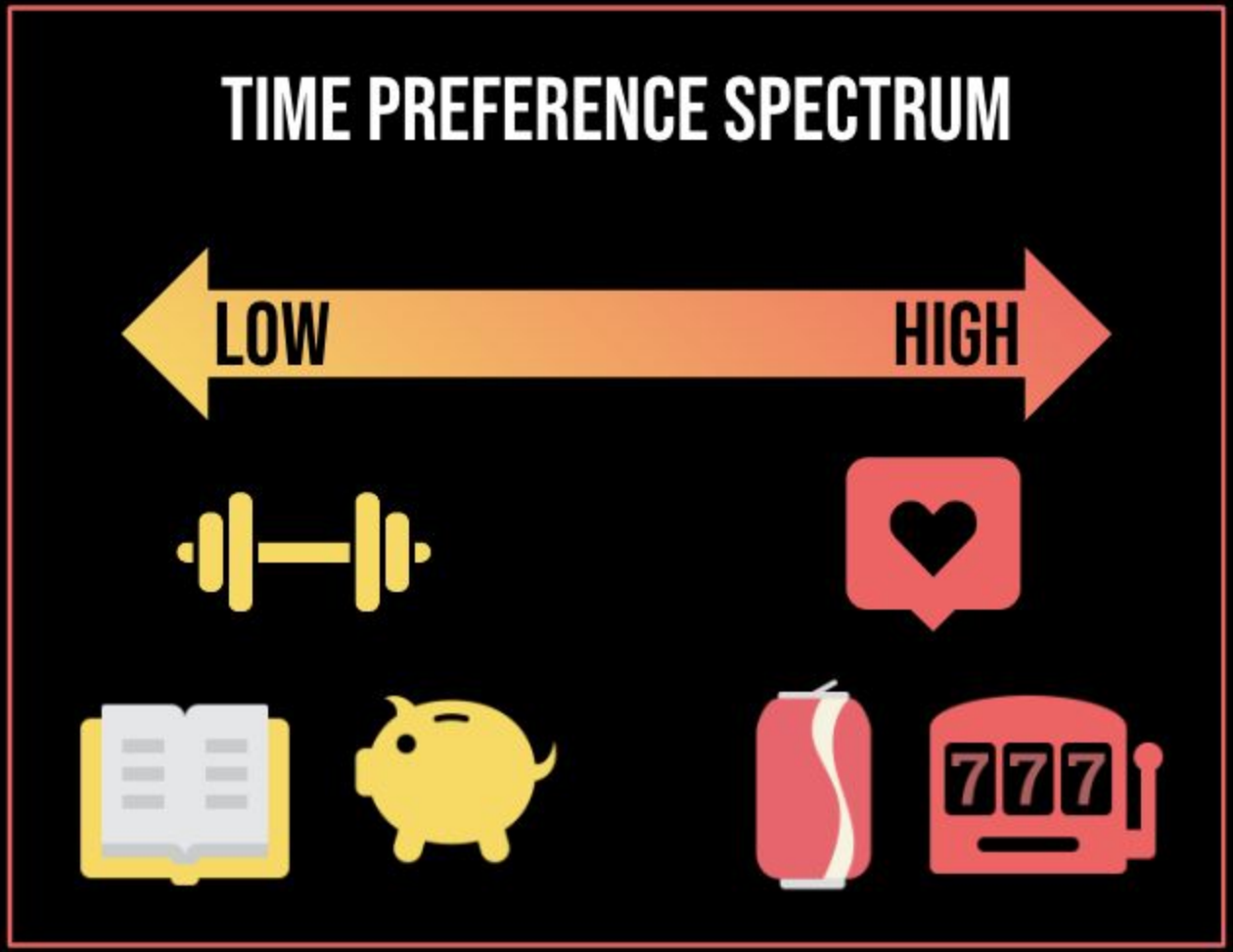
TIME PREFERENCE

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The degree to which someone values the present relative to the future

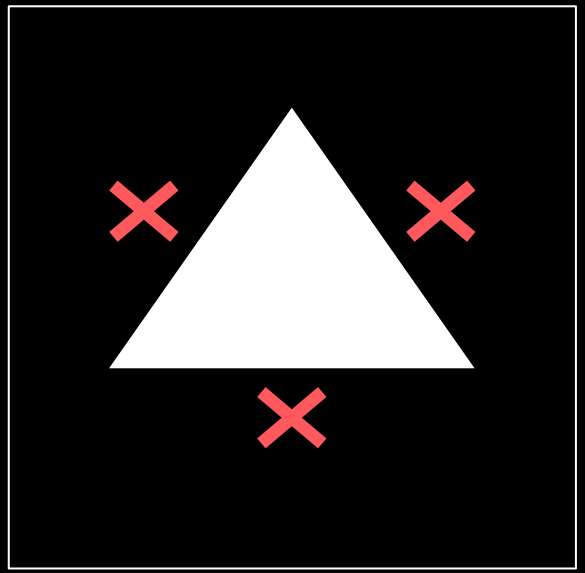
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IMPOSSIBLE TRINITY

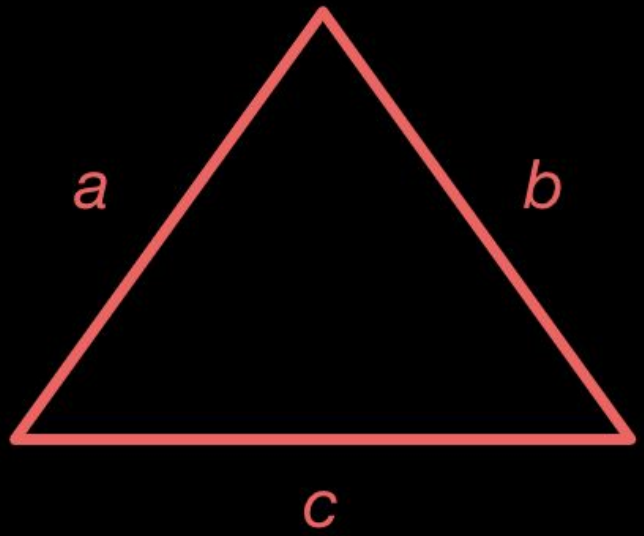
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A sovereign nation cannot simultaneously have free capital flows, independent monetary policy and a fixed exchange rate

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PICK ANY TWO



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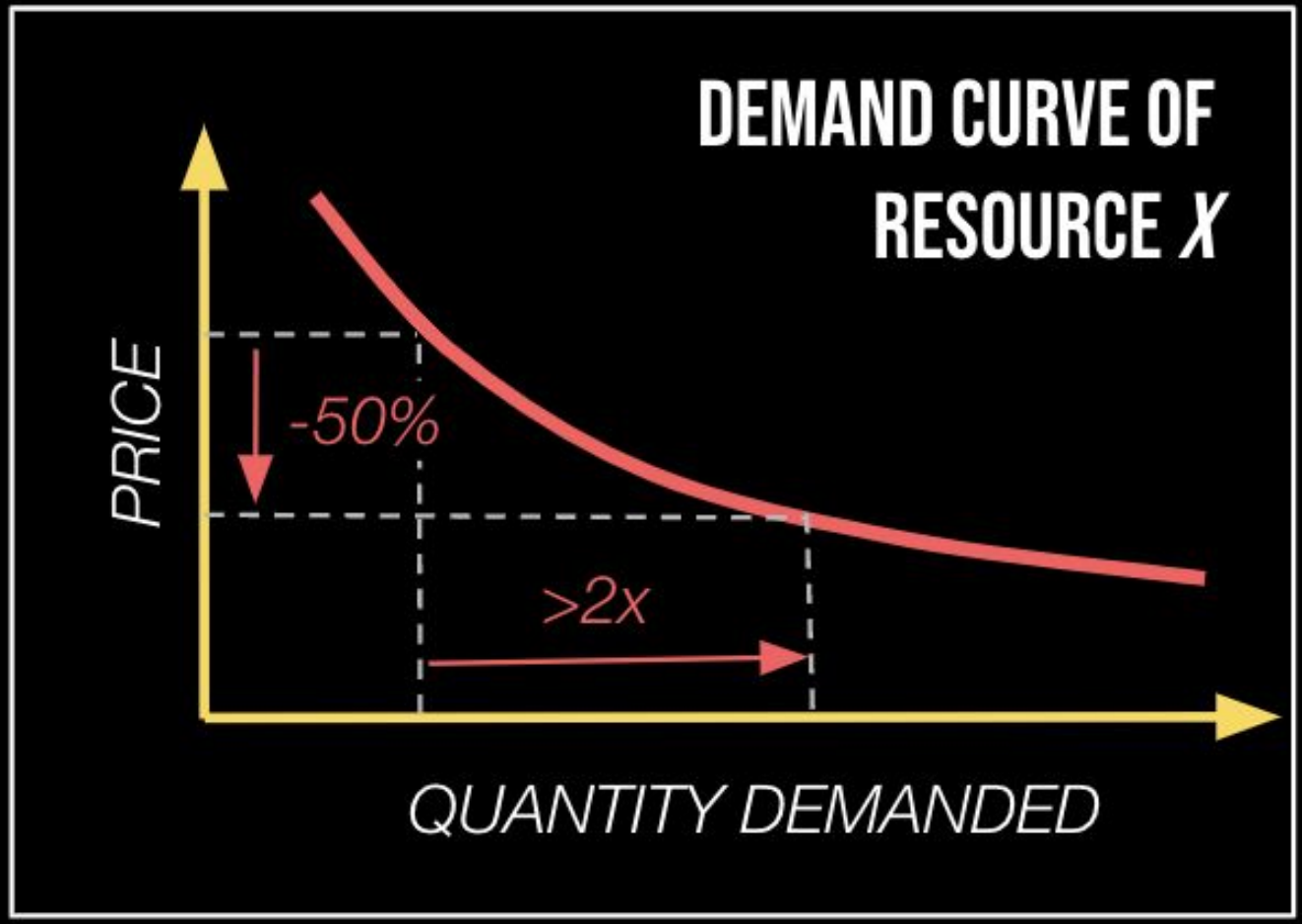
JEVONS PARADOX

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The observation that consumption of a resource increases when efficiency gains are achieved through its use

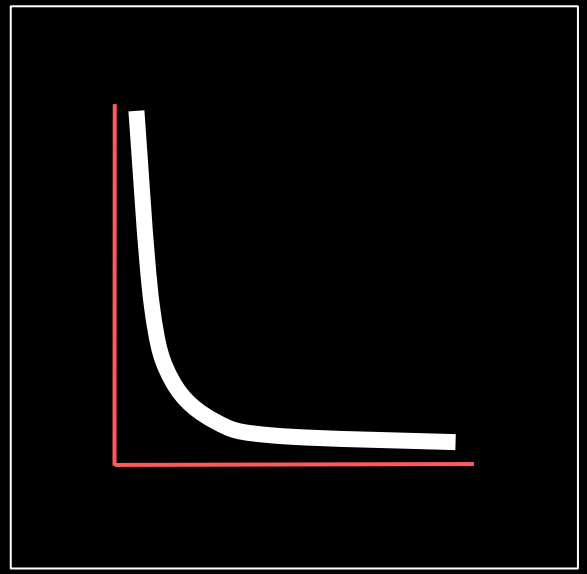
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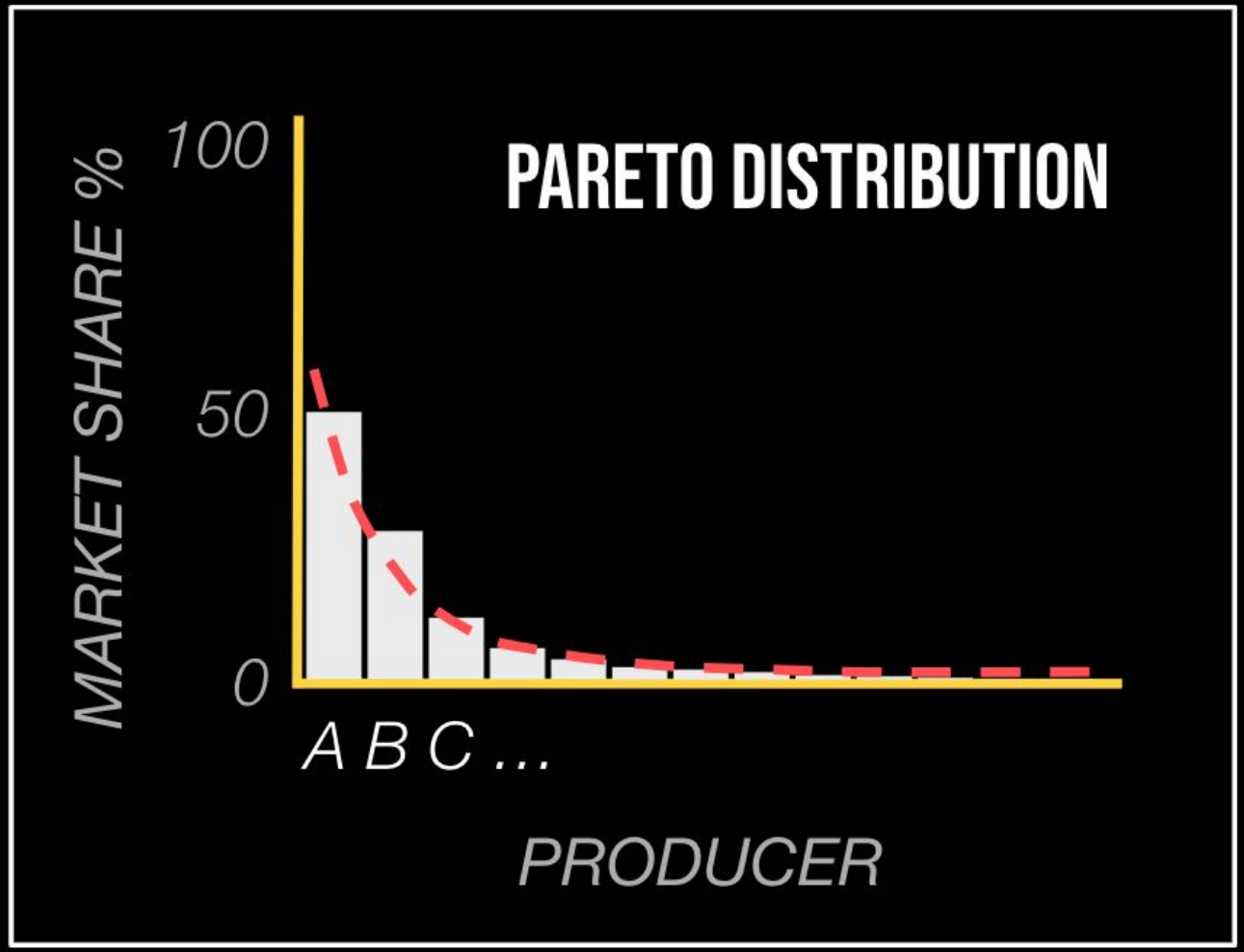
POWER LAWS

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Relationships between two quantities, where changes to one leads to a proportional relative change to the other.

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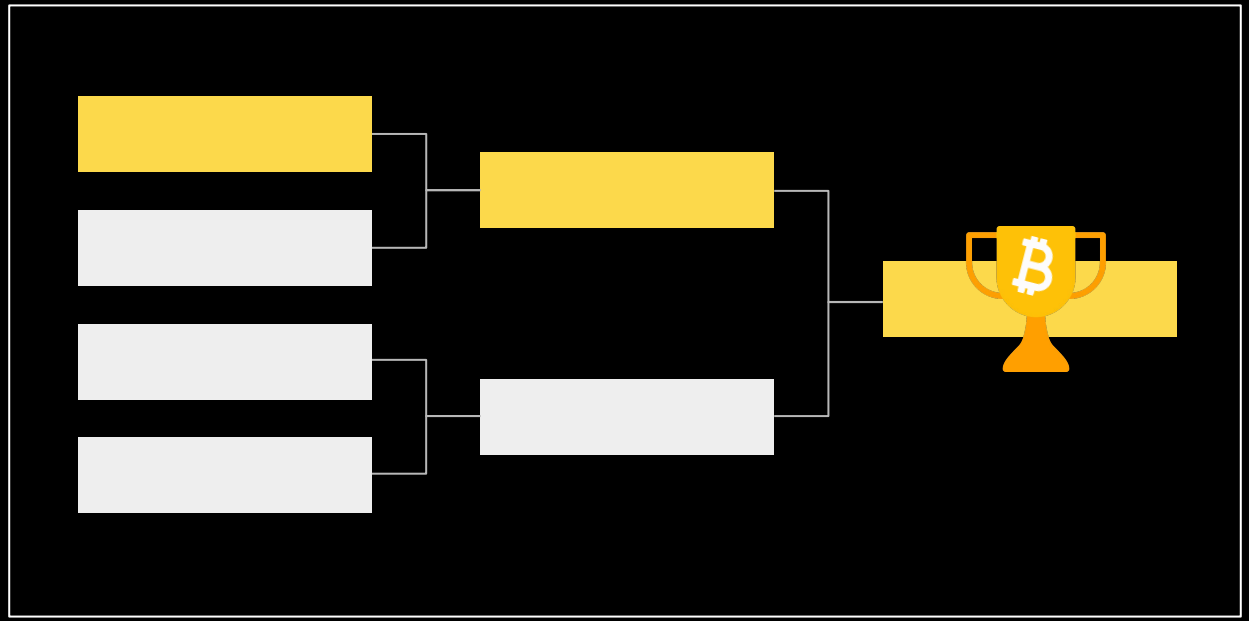
WINNER-TAKE-ALL MARKETS

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Competitions where the single best performer captures the entire reward on offer

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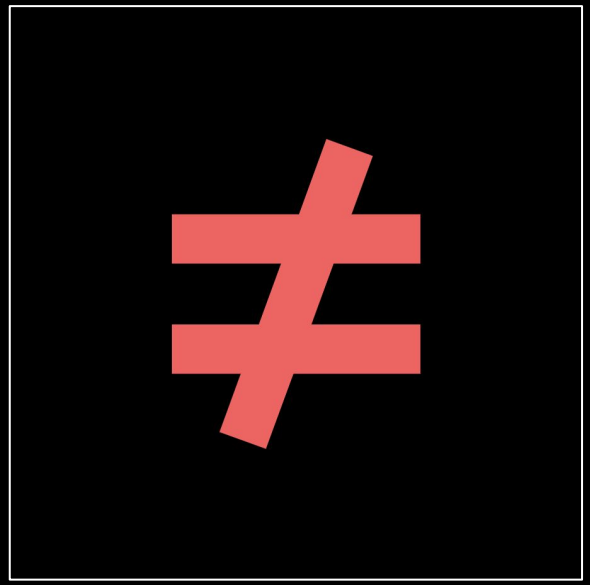
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“Money is a network.
Some networks are singular
(i.e., winner-takes-all).
Money is such a network.”

—GIGI

UNIT BIAS

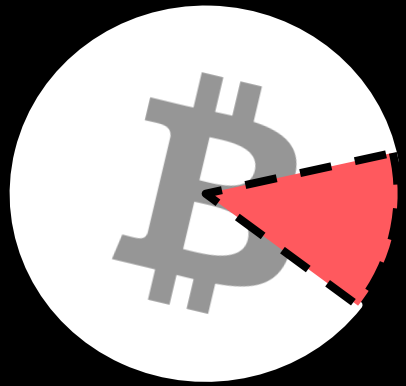
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The faulty reasoning that a single unit must be the appropriate amount from which to make assessments and comparisons

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YOU CAN BUY A



FRACTION OF A BITCOIN

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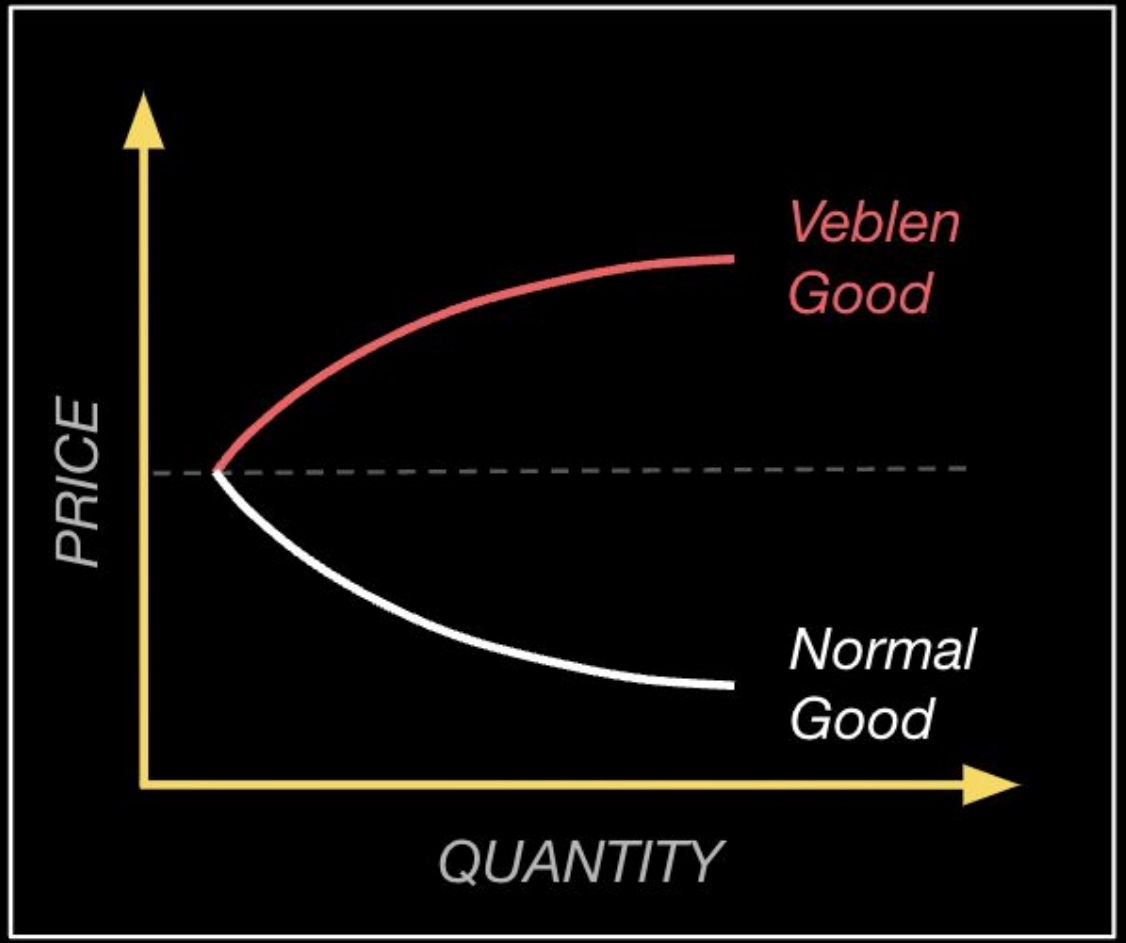
VEBLEN GOOD

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Types of goods where the quantity demanded increases as the price increases

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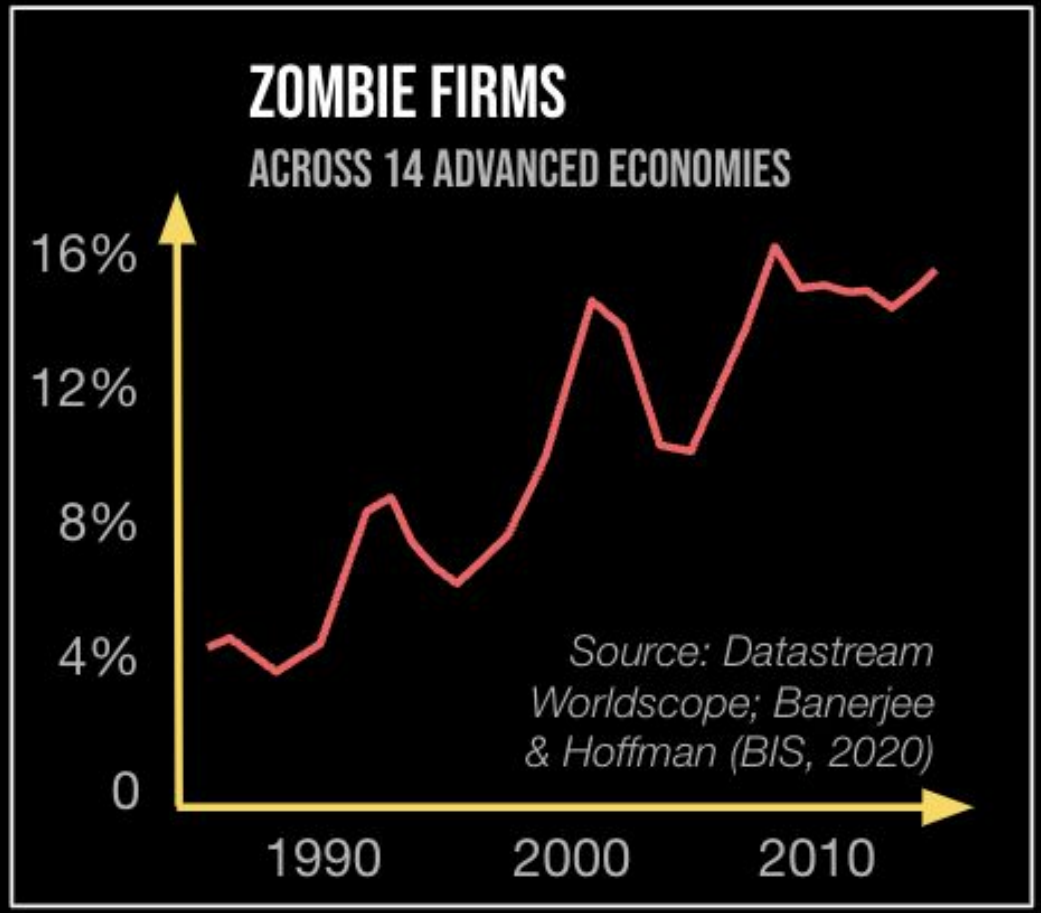
MALINVESTMENT

The Bitcoin Handbook



Distorted price signals cause a misallocation of capital towards less productive uses than would otherwise be chosen under a free market

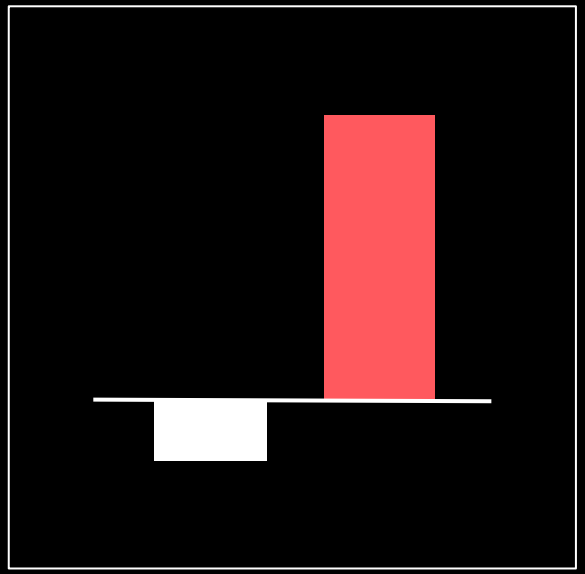
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ZOMBIE FIRMS
ACROSS 14 ADVANCED ECONOMIES
16%
12%
8%
4%
0
1990
2000
2010
Source: Datastream Worldscope; Banerjee & Hoffman (BIS, 2020)

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ASYMMETRIC PAYOFF

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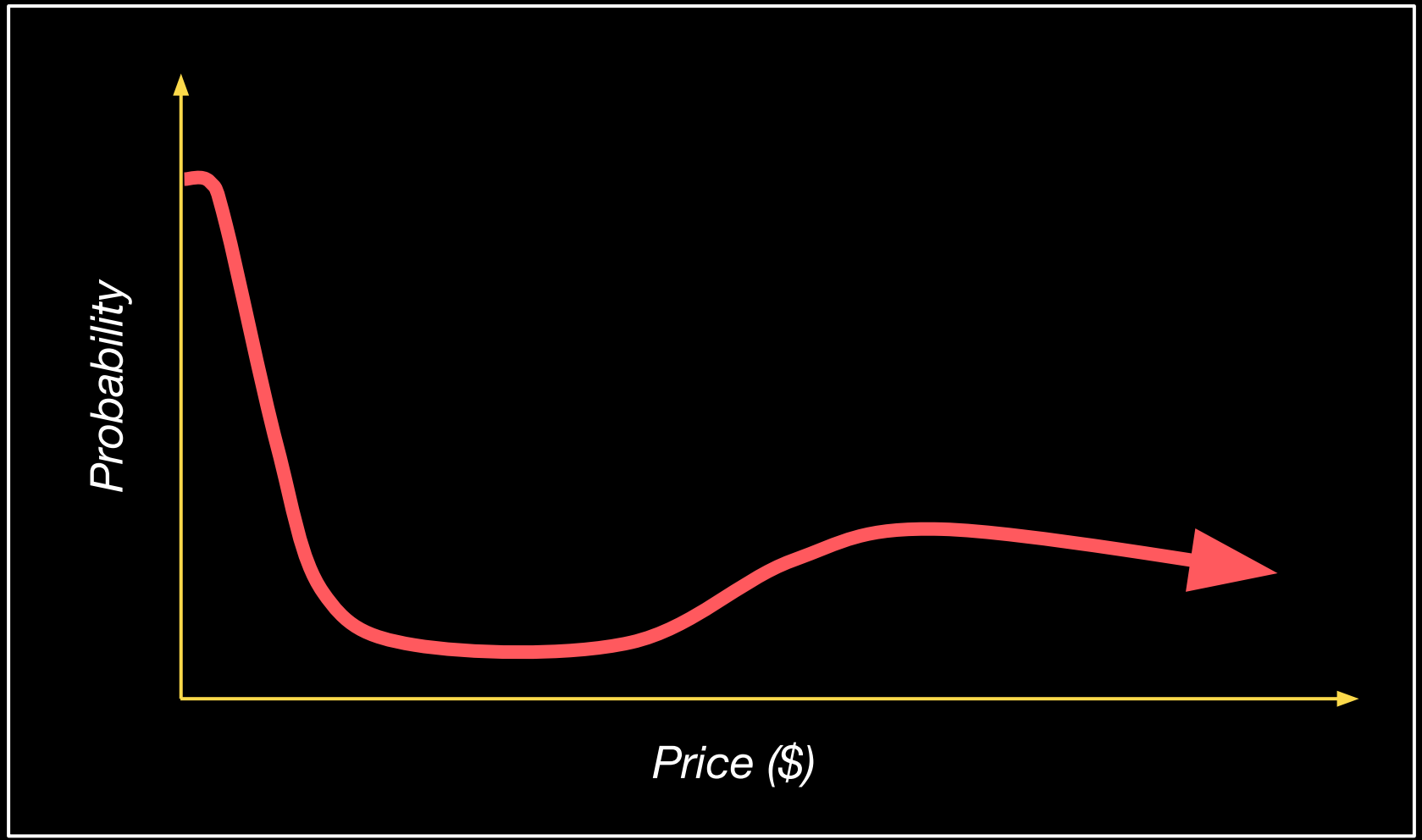


Investment decisions where the potential upside is disproportionately greater than the downside risk

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“We can debate whether or not bitcoin is going to be a big deal in the future. But I don’t think we can debate that if it is it’s going to have an enormous right tail.”

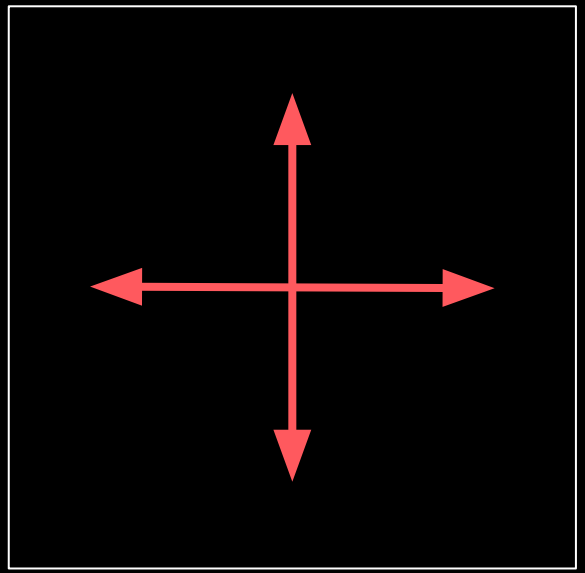
—ROSS STEVENS



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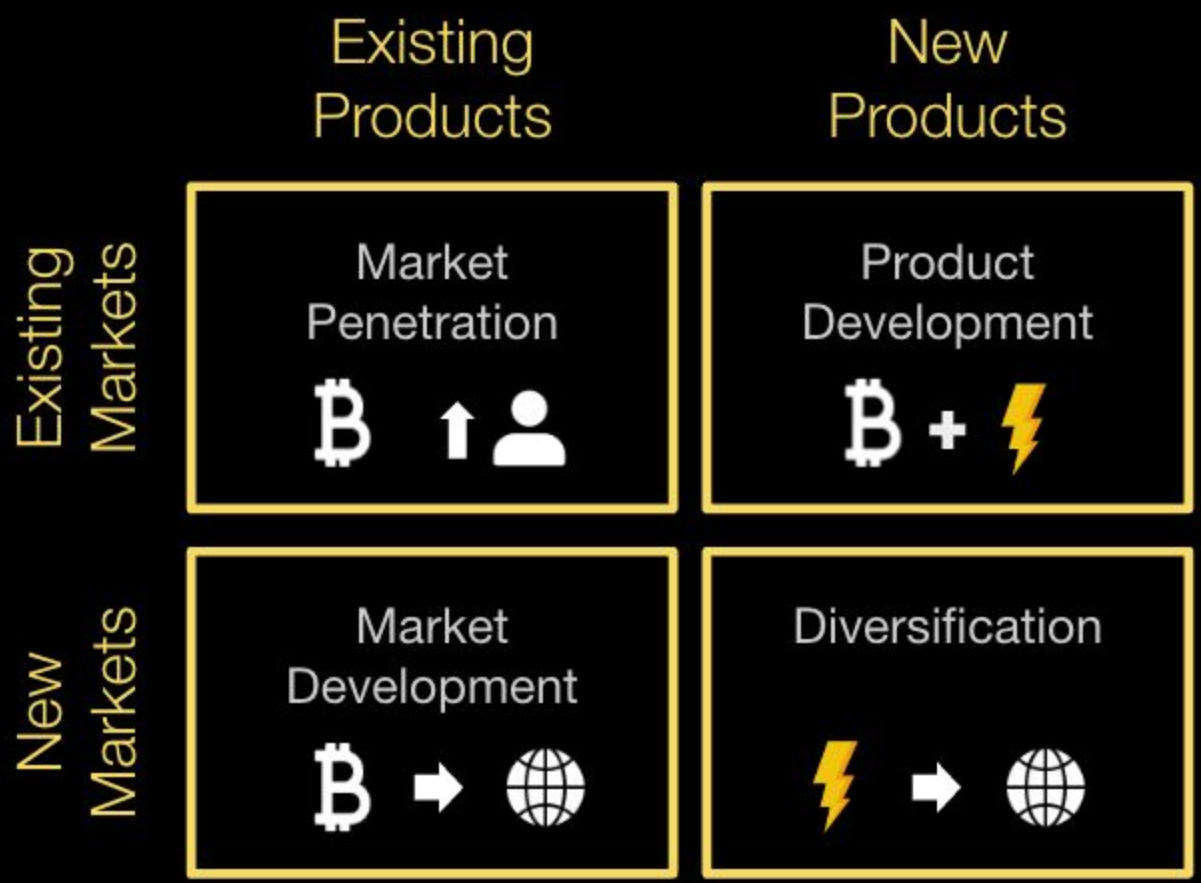
ANSOFF MATRIX

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A strategic framework for growth opportunities of a product across lines and markets

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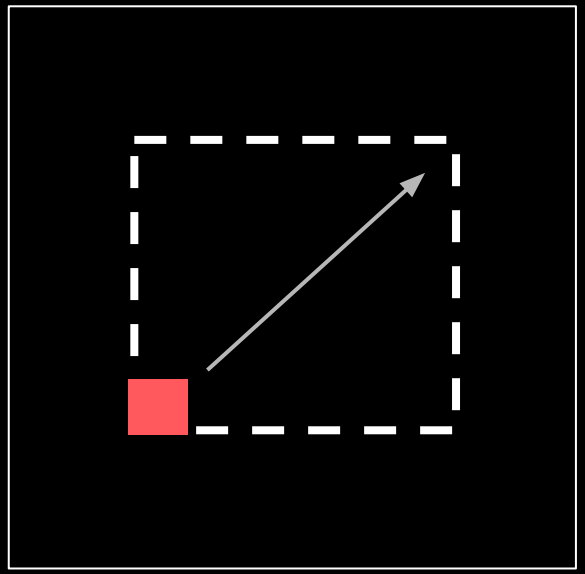

Existing Products
New Products
Existing Markets
New Markets
Market Penetration
Product Development
Market Development
Diversification

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Technology & Systems

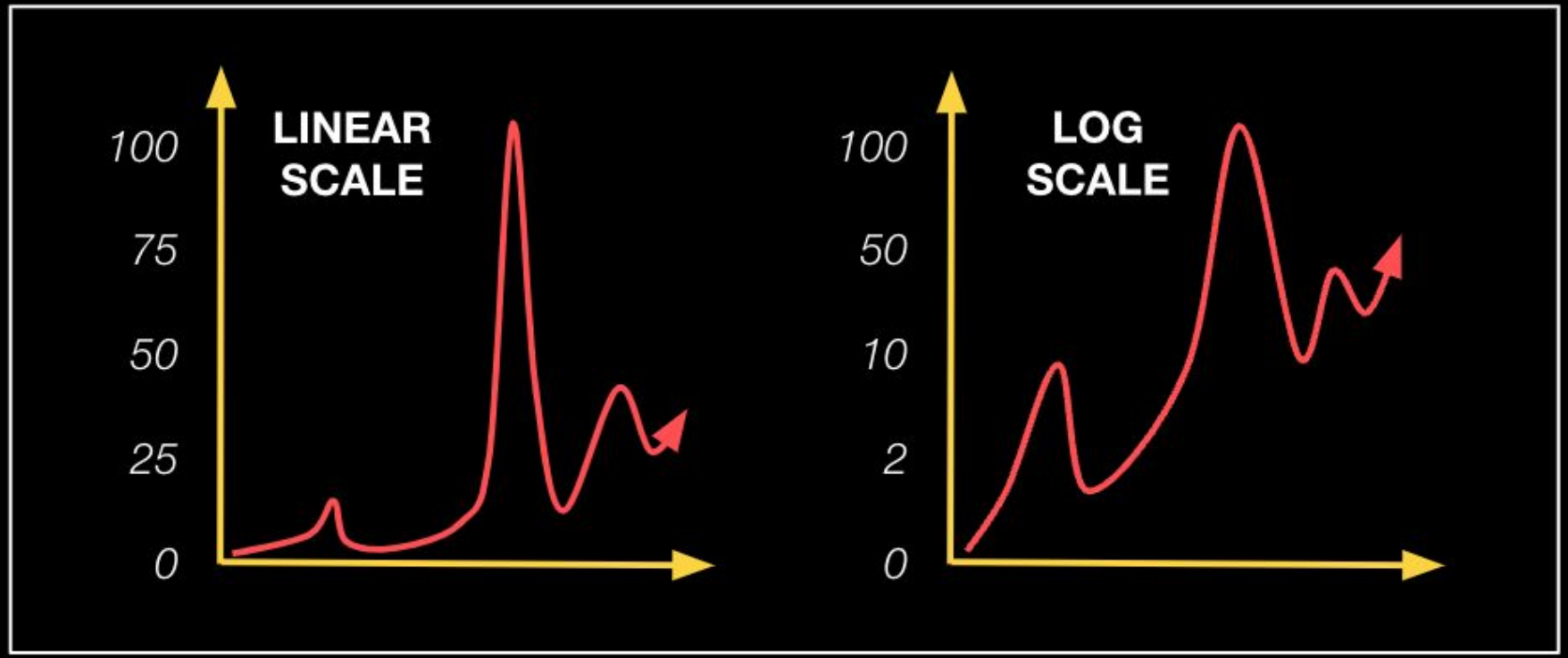
ORDERS OF MAGNITUDE

The Bitcoin Handbook



When comparing two measurements in linear terms in infeasible, multiples of ten can provide a more intuitive scale

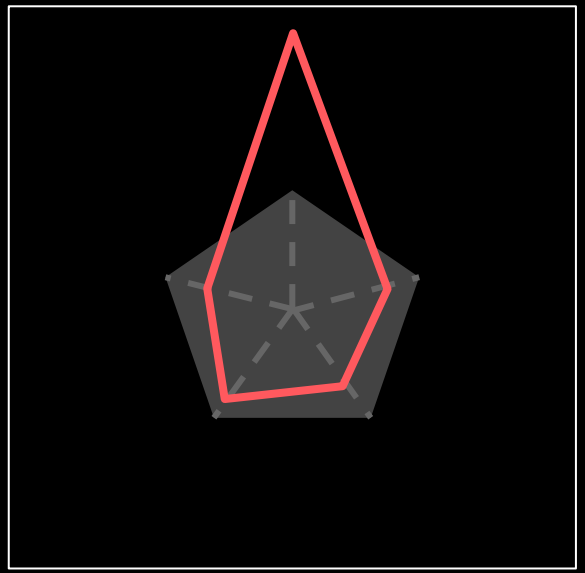
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10X IMPROVEMENT RULE

BITCOIN
ESSENTIAL CONCEPTS



New technologies must offer a benefit ten times greater than a predecessor or substitutes to gain widespread adoption

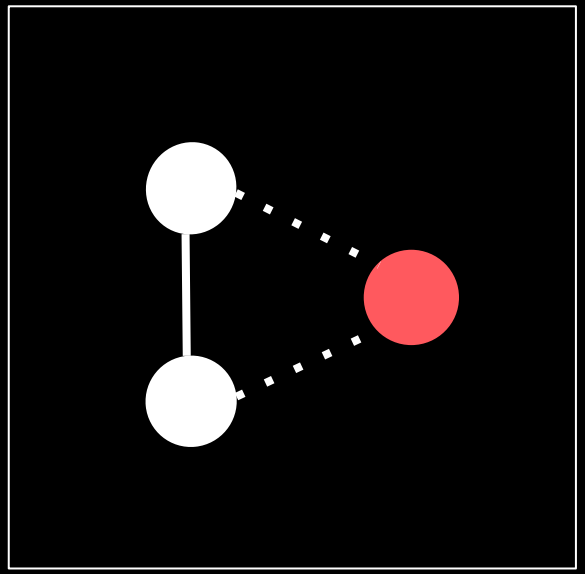
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*“As a good rule of thumb, proprietary technology must be **at least 10 times better** than its closest substitute in some important dimension to lead to a real monopolistic advantage.”*

—PETER THIEL

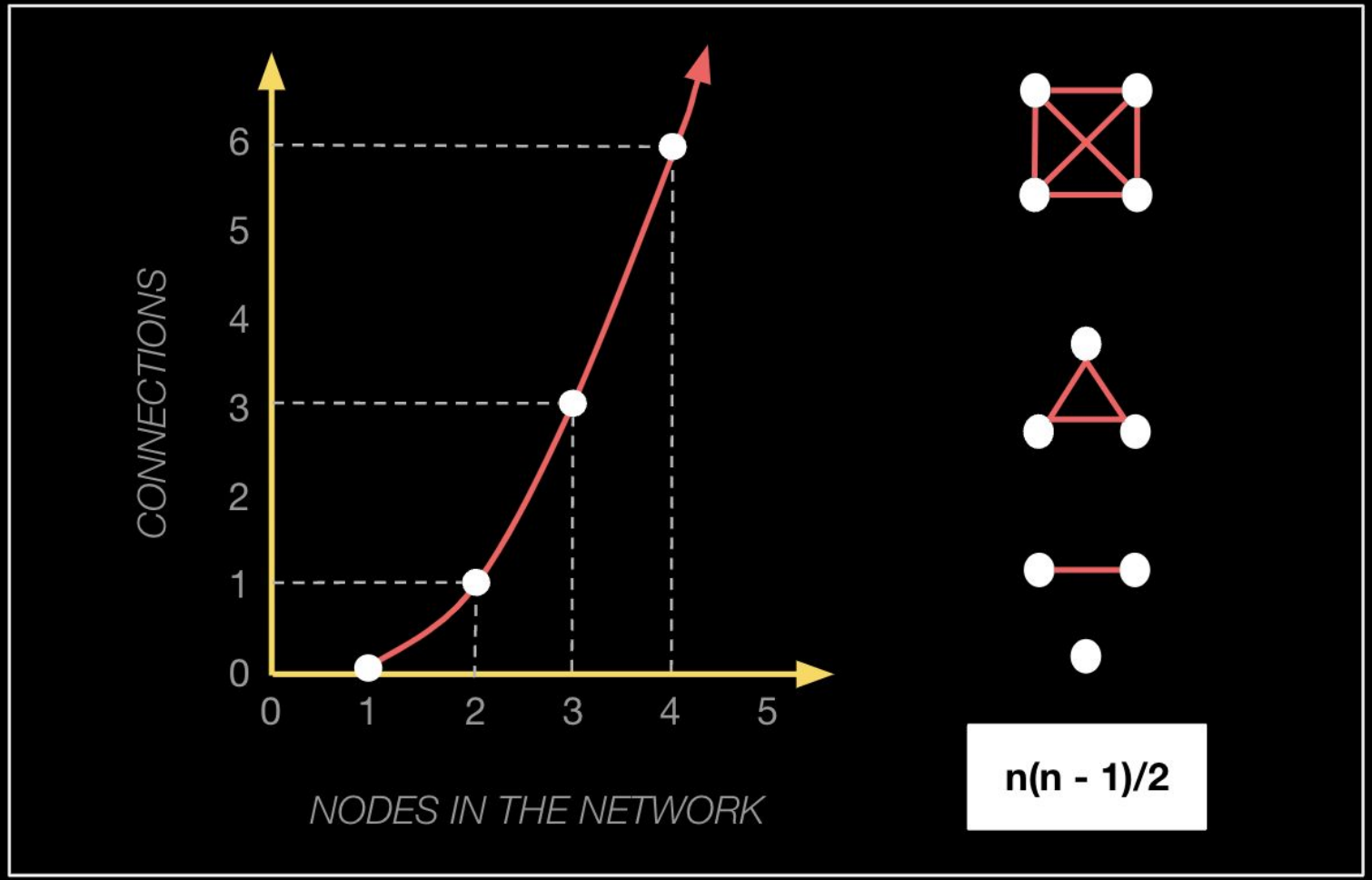
NETWORK EFFECTS

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The phenomena whereby each additional user to a network adds disproportionately more value and utility

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CHURN

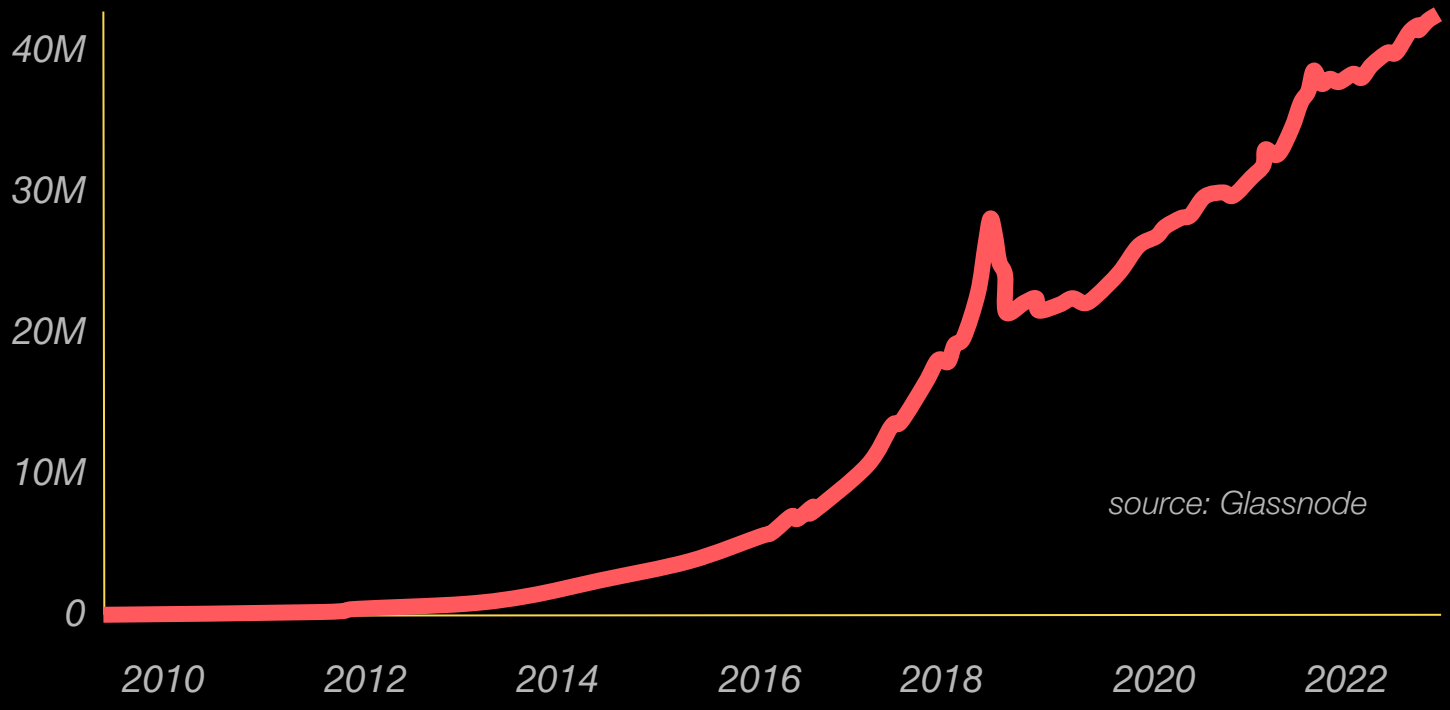
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The rate of user attrition, often expressed as a percentage

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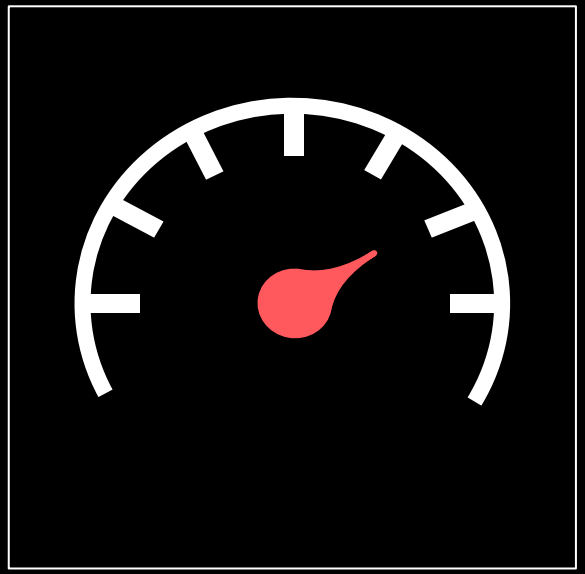
NON-ZERO BITCOIN ADDRESSES



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ACCELERATING RETURNS

The Bitcoin Handbook



The rate of technological change is exponential, resulting in ever-increasing innovation and disruption

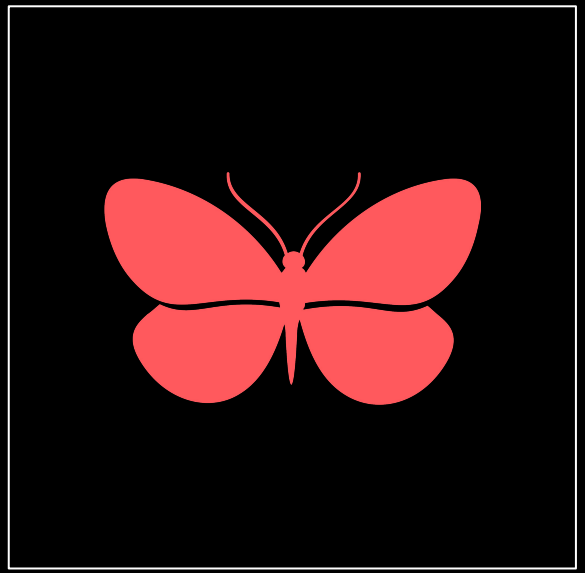
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TECHNOLOGICAL CHANGE
You are here
Actual
Expectation
TIME

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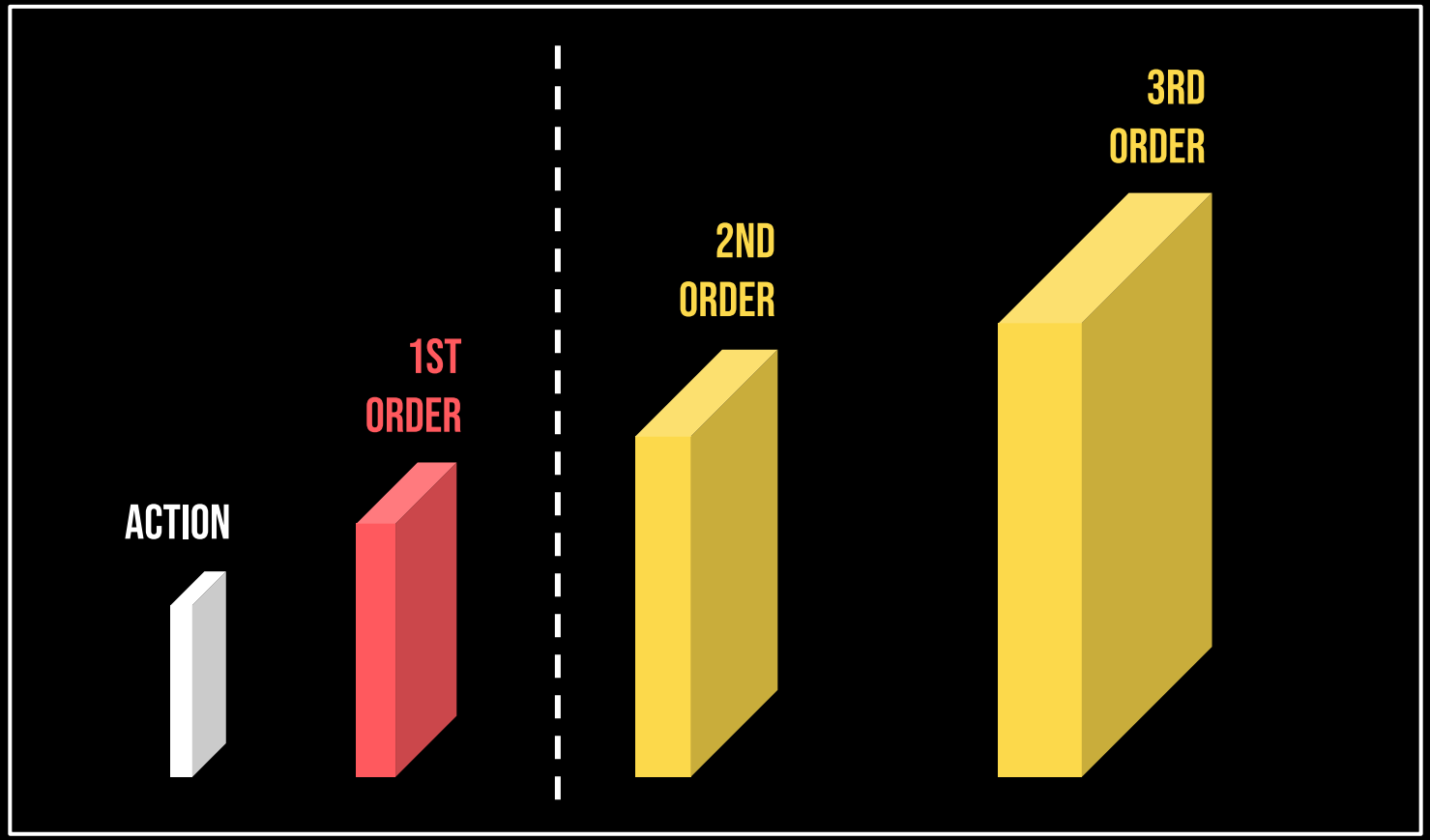
HIGHER-ORDER EFFECTS

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Actions have consequences.
And those consequences
have consequences.

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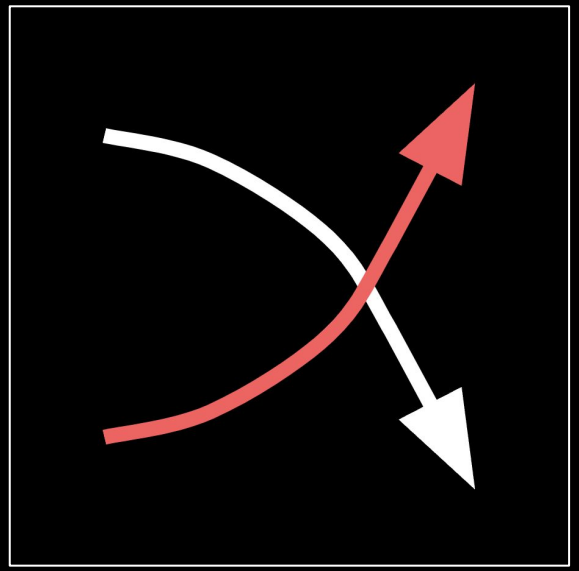
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“Many people look at first order effects, thinking in the short term. Conversely, not spending time considering the 2nd and 3rd order effects of actions—which leads them to put too much trust in a system that will ultimately fail.”

—JEFF BOOTH

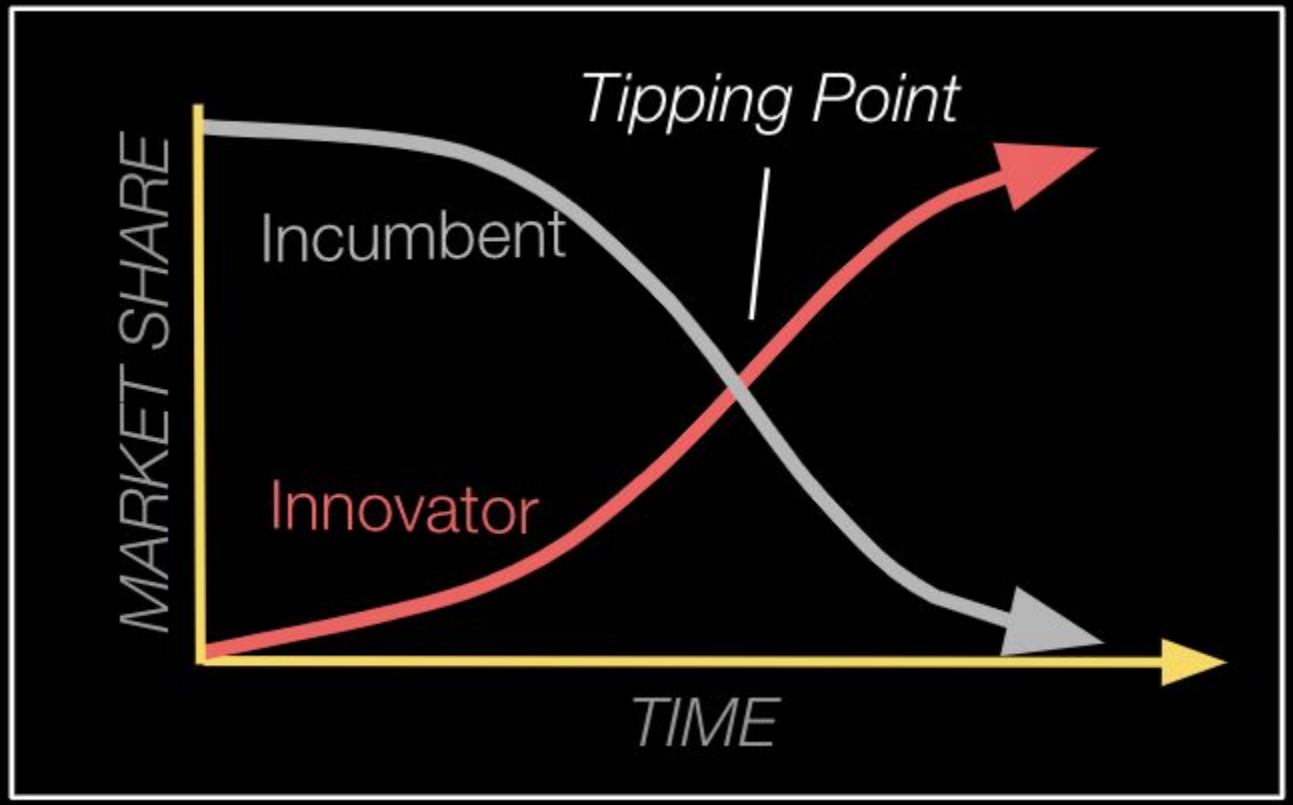
CREATIVE DESTRUCTION

The Bitcoin Handbook



The cycle of entrepreneurs out-innovating and obsoleting incumbents, under a free-market

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FEEDBACK LOOPS

The Bitcoin Handbook

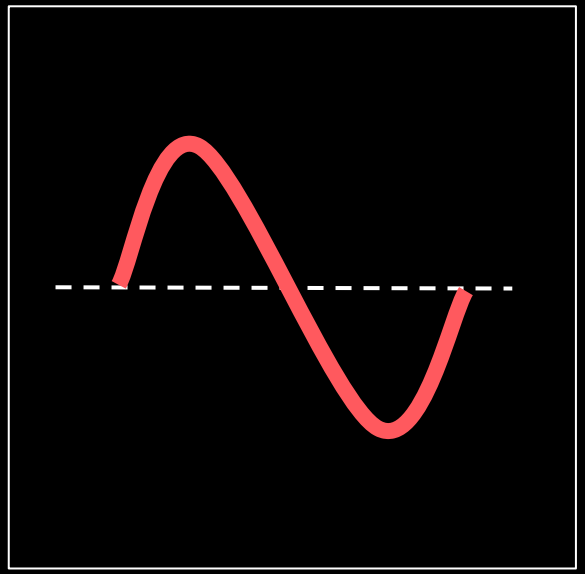


A system that reacts to its environment and the resultant output gets re-incorporated as an input

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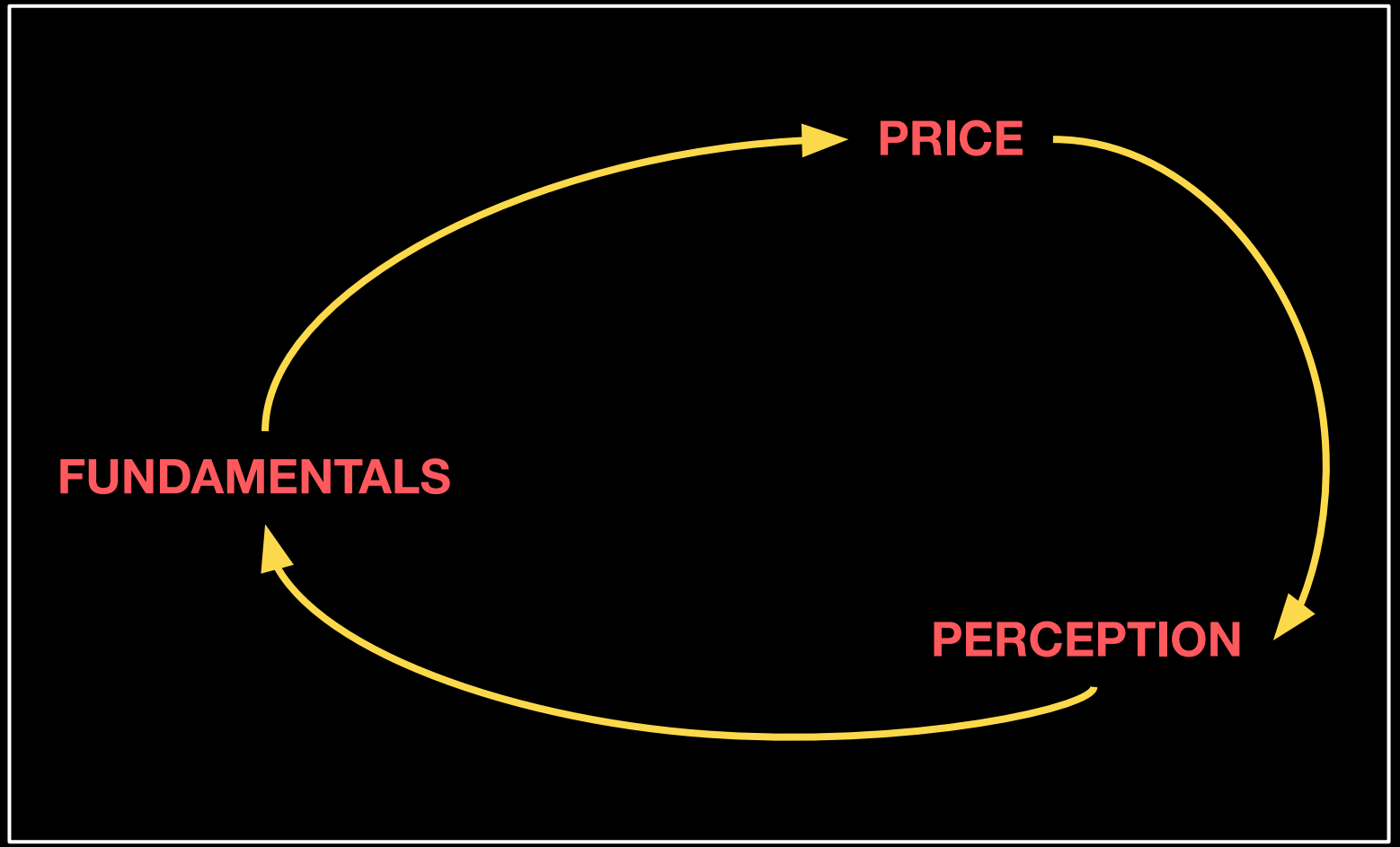
REFLEXIVITY

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The circular relationship between market participants and market sentiment, whereby attributing the respective impact of either is difficult

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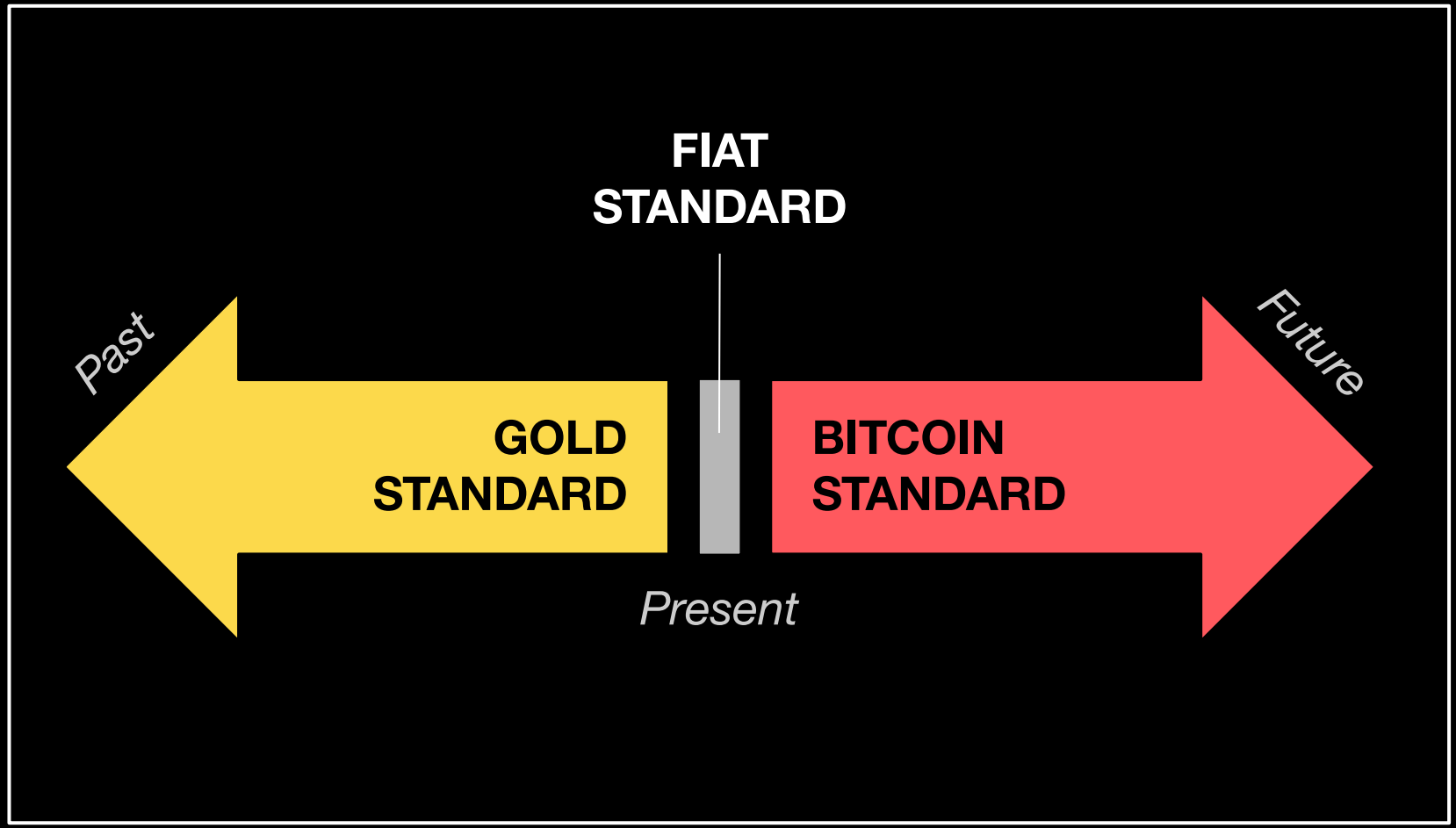
RELATIVITY

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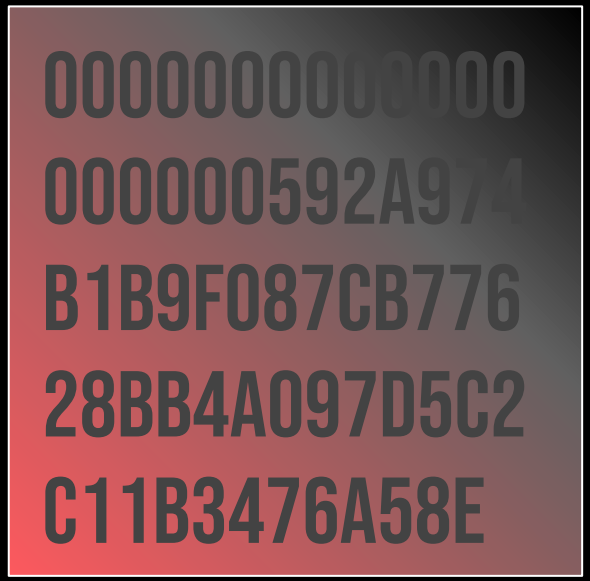
When you're a part of a system, it can be difficult to zoom out to gain perspective


FIAT
STANDARD
Past
GOLD
STANDARD
Present
BITCOIN
STANDARD
Future

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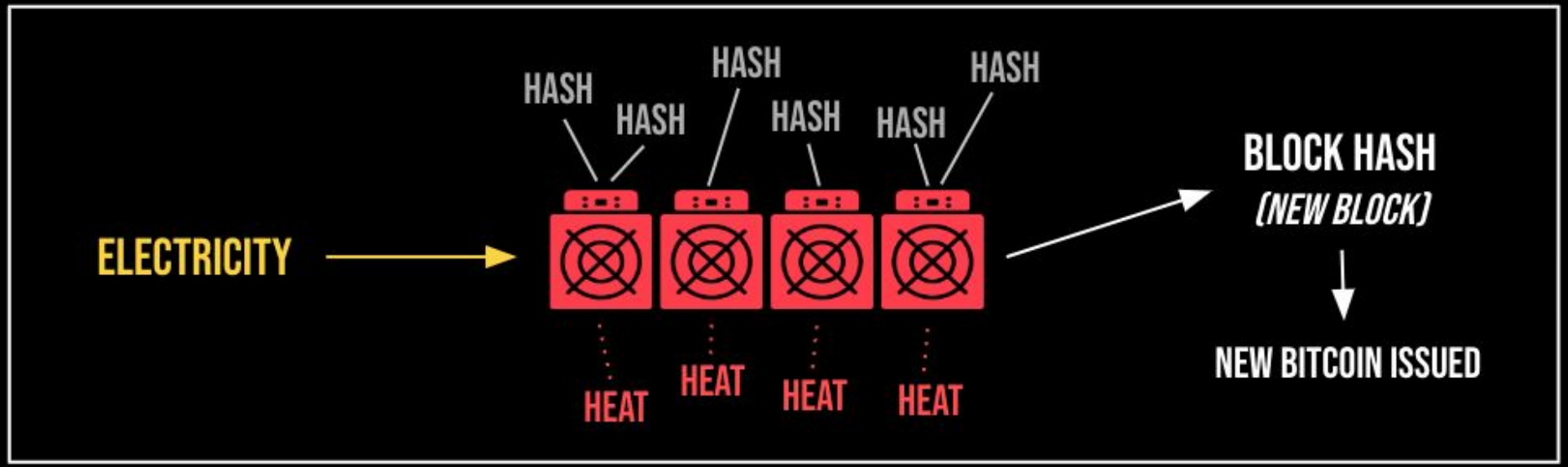
THERMODYNAMICS 1ST LAW

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Energy cannot be created nor destroyed in isolated systems. It can only be transformed


HASH
HASH
HASH
HASH
HASH
HASH
ELECTRICITY
BLOCK HASH
(NEW BLOCK)
NEW BITCOIN ISSUED
HEAT
HEAT
HEAT
HEAT

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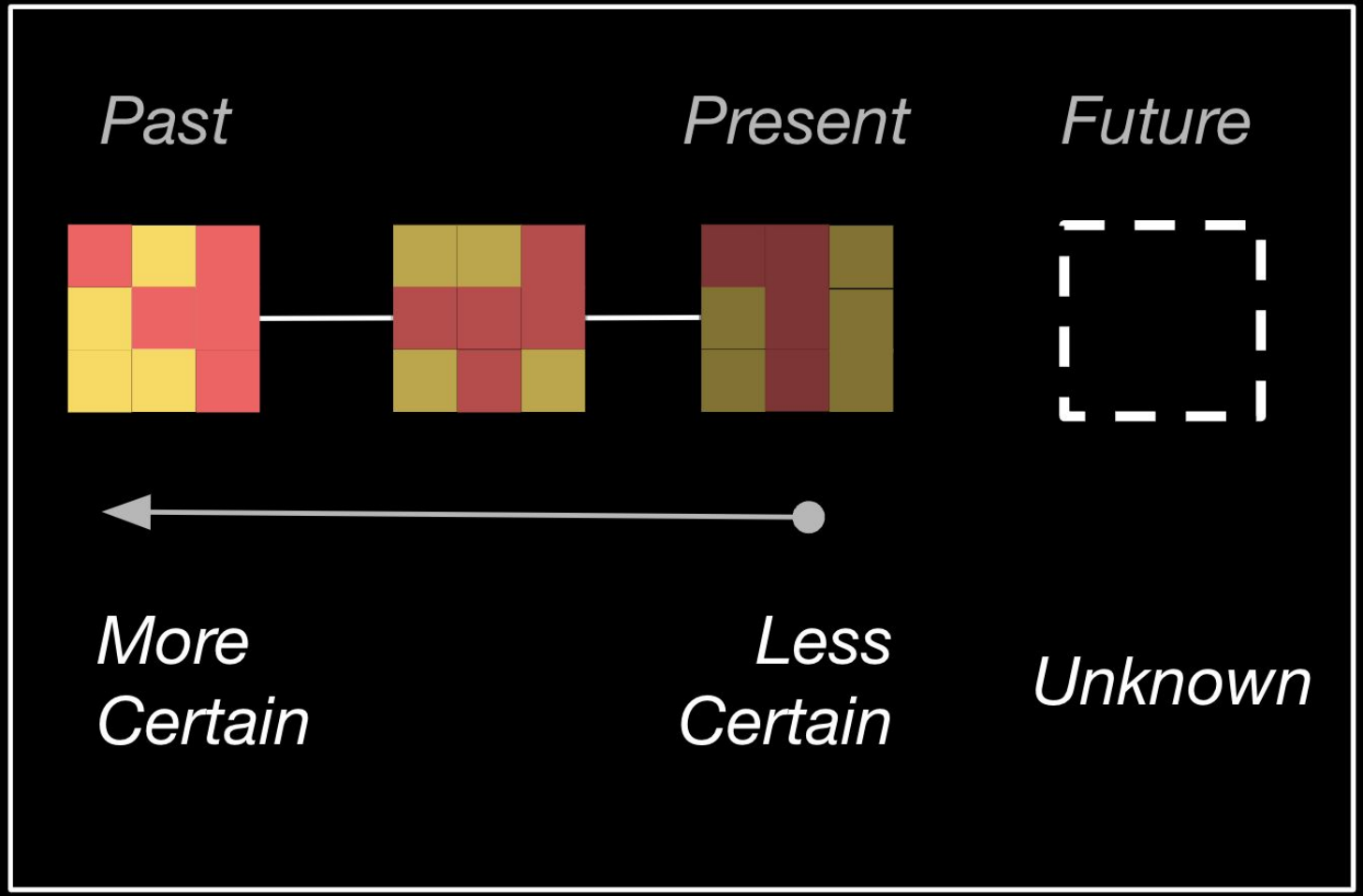
THERMODYNAMICS 2ND LAW

The Bitcoin Handbook



The entropy of the universe always increases with time.

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ARROW OF TIME

The Bitcoin Handbook

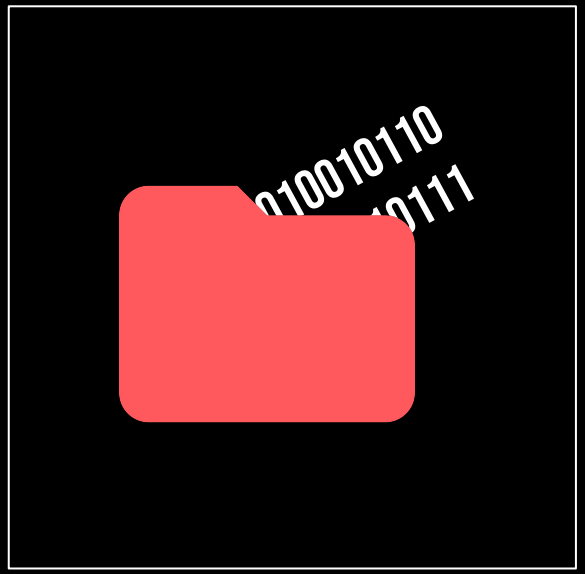


Establishing the one-way direction of time by distinguishing past from present.

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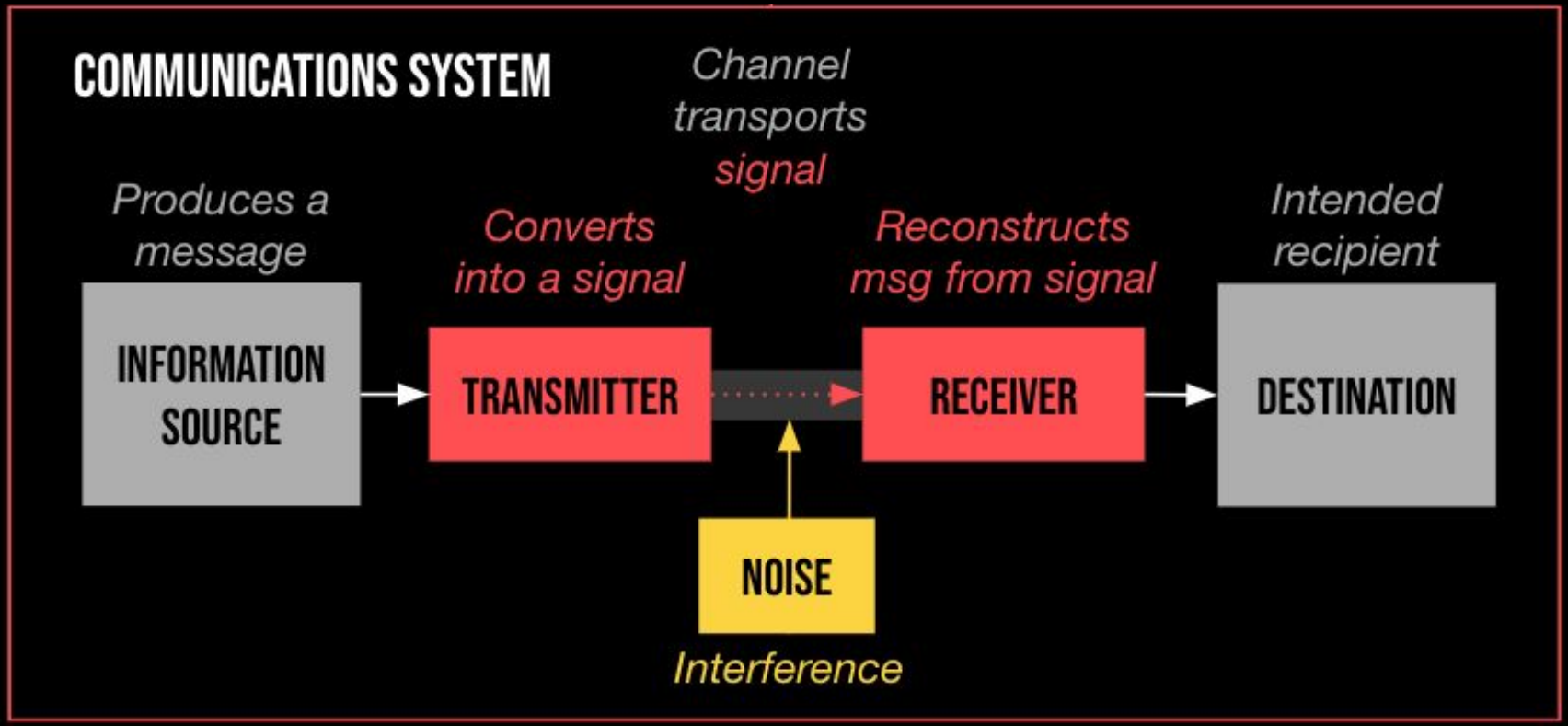
INFORMATION THEORY

The Bitcoin Handbook



How digital information is communicated, stored, and quantified.

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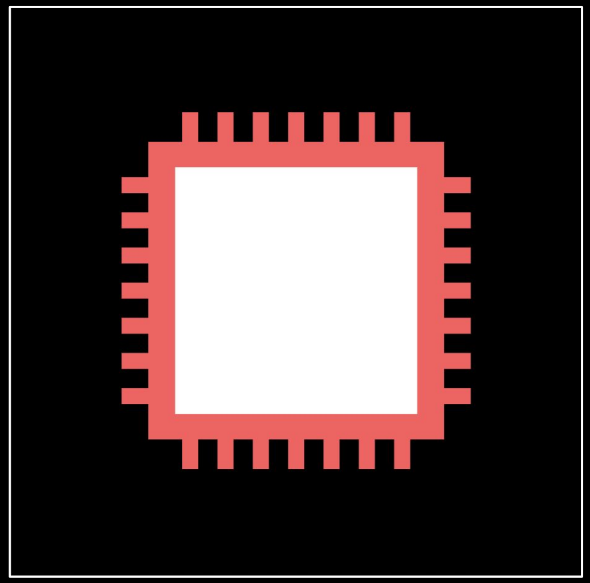
Adapted from *A Mathematical Theory of Communication* by Claude Shannon (1949).

"The fundamental problem of communication is that of reproducing at one point, either exactly or approximately, a message selected at another point."

—CLAUDE SHANNON

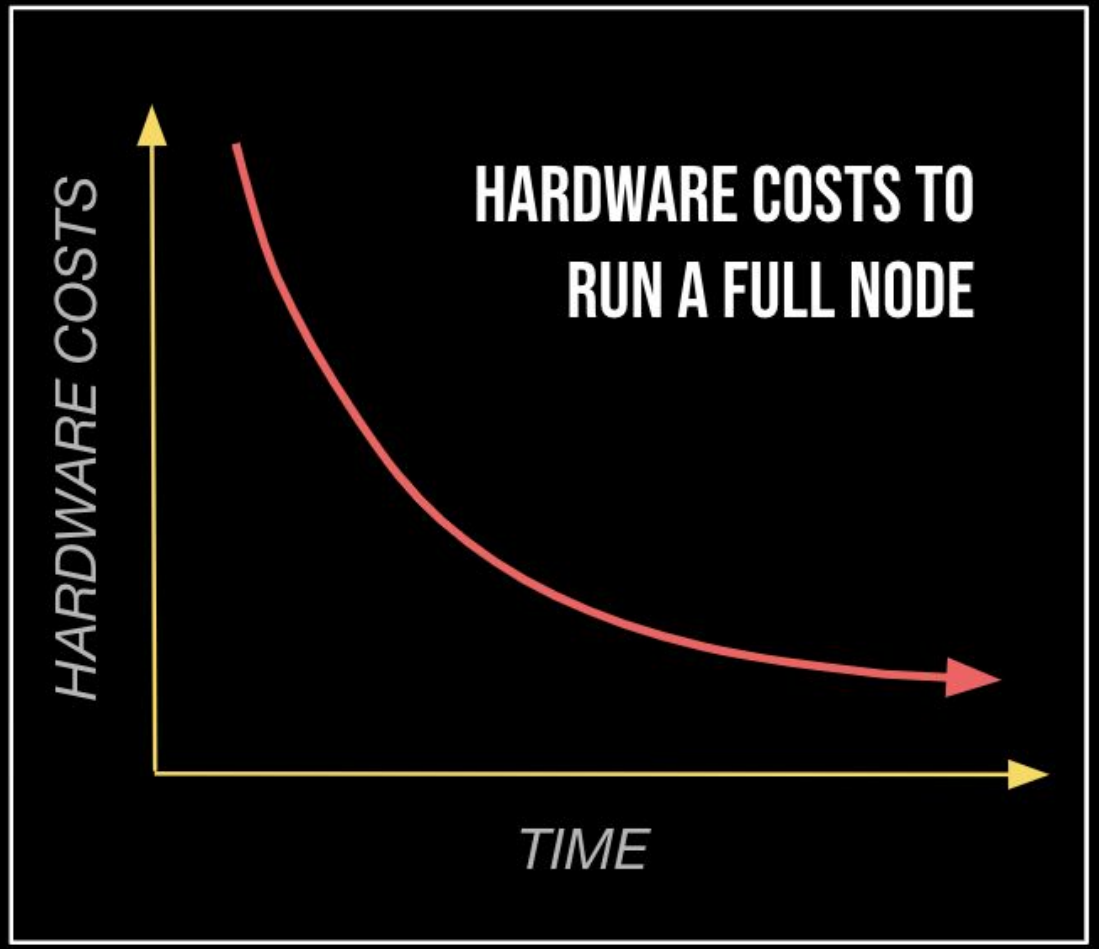
MOORE'S LAW

The Bitcoin Handbook



The observation that the number of transistors on a computer chip doubles approximately every two years

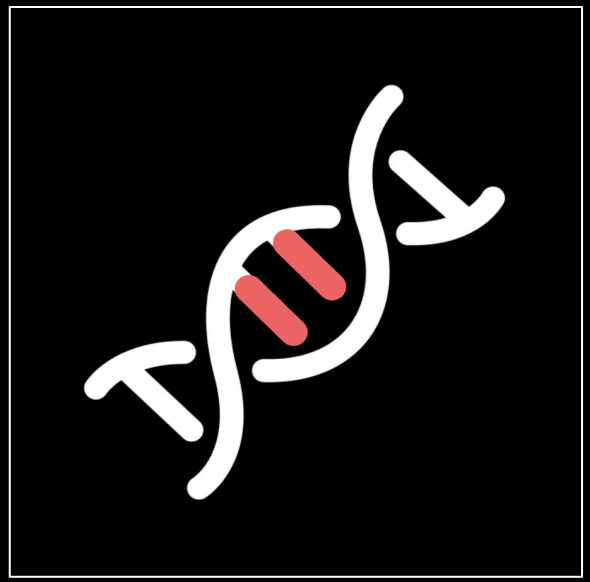
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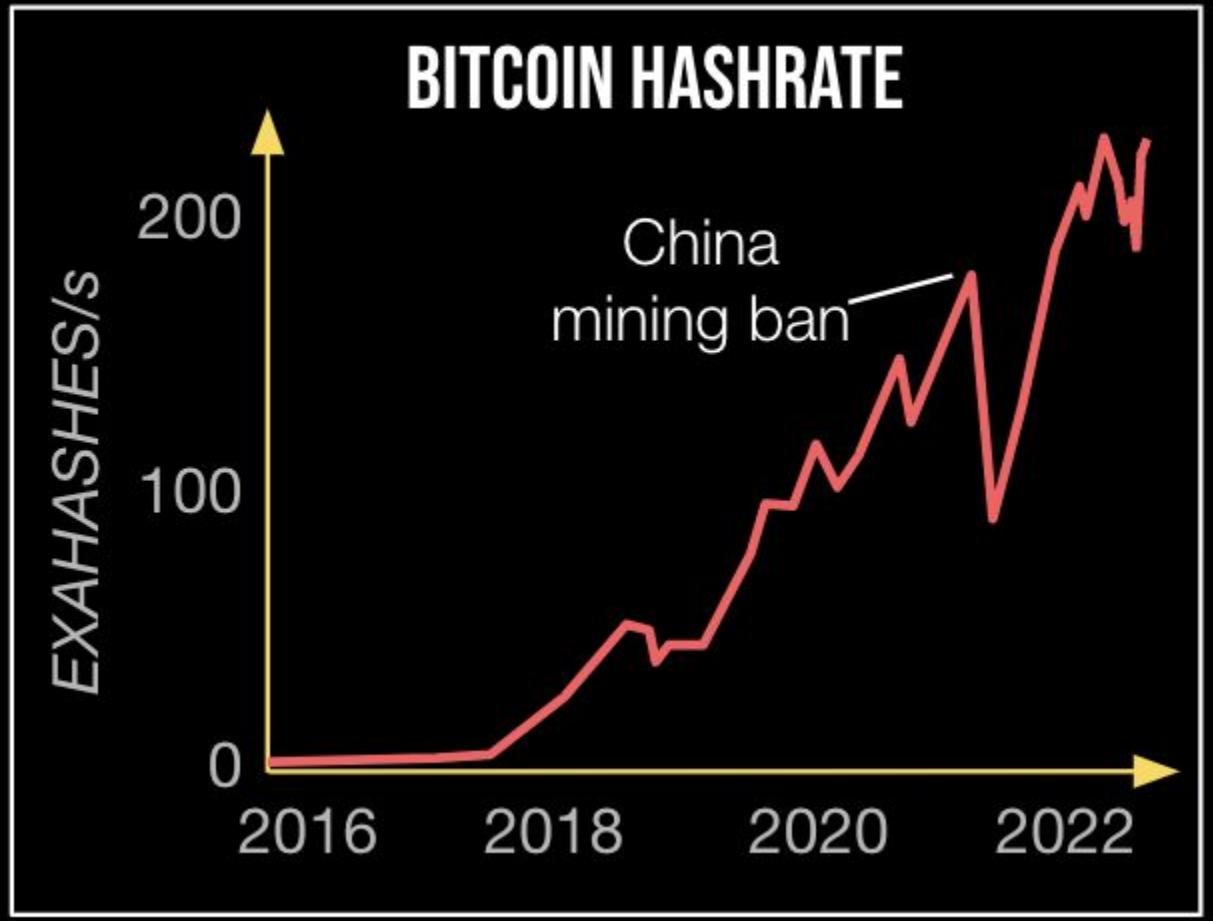
ANTIFRAGILITY

The Bitcoin Handbook



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Things that actively gain from volatility and unpredictability by using rapid feedback loops to evolve


BITCOIN HASHRATE
EXAHASHES/s
200
100
0
China
mining ban
2016
2018
2020
2022

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GALL'S LAW

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Incremental improvements made to a functional system is a superior strategy to building a complex system from scratch

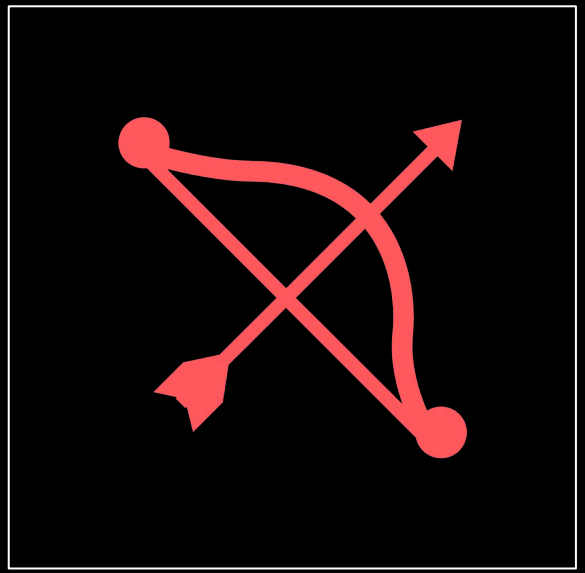
*“Bitcoin is too important to follow the Silicon Valley mantra of move fast and break things. Instead, it’s move slowly and don’t break **anything**.*

If a global financial system is to be built on a decentralized monetary system, the foundation must be protected at all cost.”

—PARKER LEWIS

ATTACK SURFACE

The Bitcoin Handbook



The sum of all potential vulnerabilities from all potential access points

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```
CAmount GetBlockSubsidy(int nHeight, const
Consensus::Params& consensusParams)
{
    int halvings = nHeight /
consensusParams.nSubsidyHalvingInterval;
    // Force block reward to zero when right shift is
undefined.
    if (halvings >= 64)
        return 0;

    CAmount nSubsidy = 50 * COIN;
    // Subsidy is cut in half every 210,000 blocks which
will occur approximately every 4 years.
    nSubsidy >>= halvings;
    return nSubsidy;
}
```

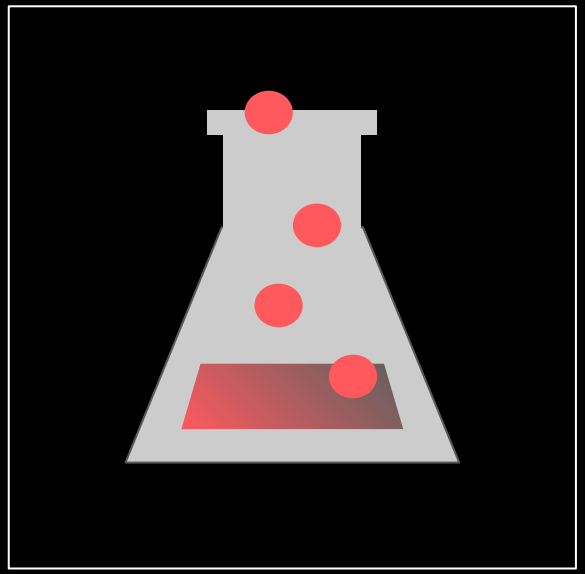
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“Being open source means anyone can independently review the code. If it was closed source, nobody could verify the security.”

—SATOSHI NAKAMOTO

CATALYSTS

The Bitcoin Handbook

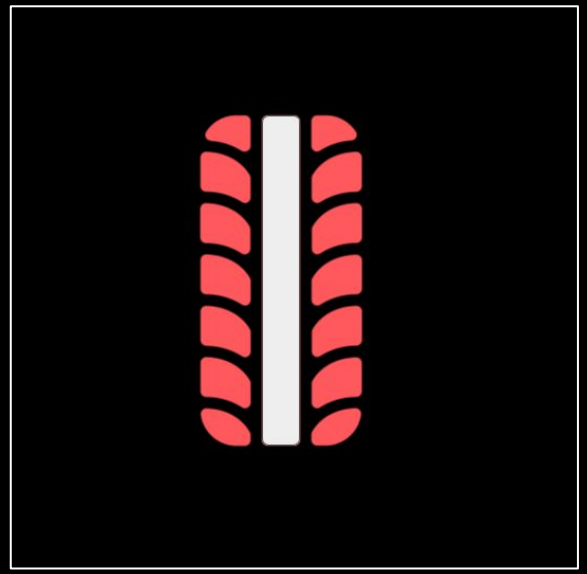


Kick-starts a reaction
but isn't itself a reactant

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FRICTION

The Bitcoin Handbook



A force that opposes the relative movement of something

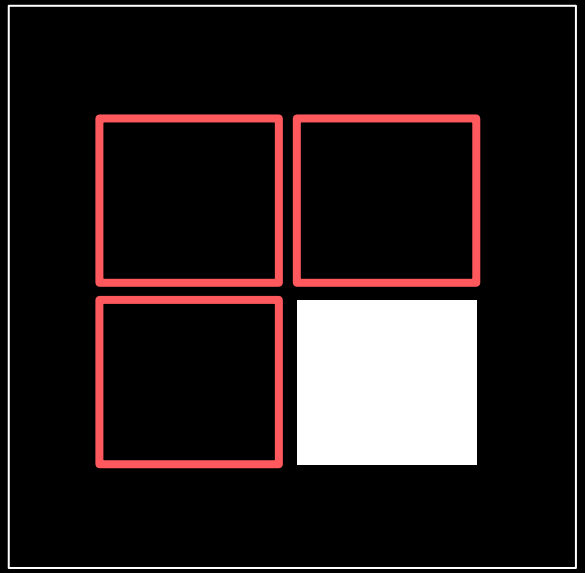
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Psychology

PRISONER'S DILEMMA

The Bitcoin Handbook



Using math to determine if it's better to cooperate or compete in multiplayer games

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Fidelity DIGITAL ASSETS℠

“If bitcoin adoption increases, the countries that secure some bitcoin today will be better off competitively than their peers. Therefore, even if other countries do not believe in the investment thesis.., they will be forced to acquire some as a form of insurance.”

—CHRIS KUIPER & JACK NEUREUTER

JURISDICTIONAL ARBITRAGE

The Bitcoin Handbook



Taking advantage of discrepancies between competing legal jurisdictions

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SOCIAL PROOF

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In uncertain situations we look to those we deem knowledgeable or competent for guidance

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“Paul Tudor Jones validates it for other hedge fund managers, hedge fund managers validate it for sovereign-wealth funds, sovereign-wealth funds will validate it for central banks.”

—NAVAL RAVIKANT

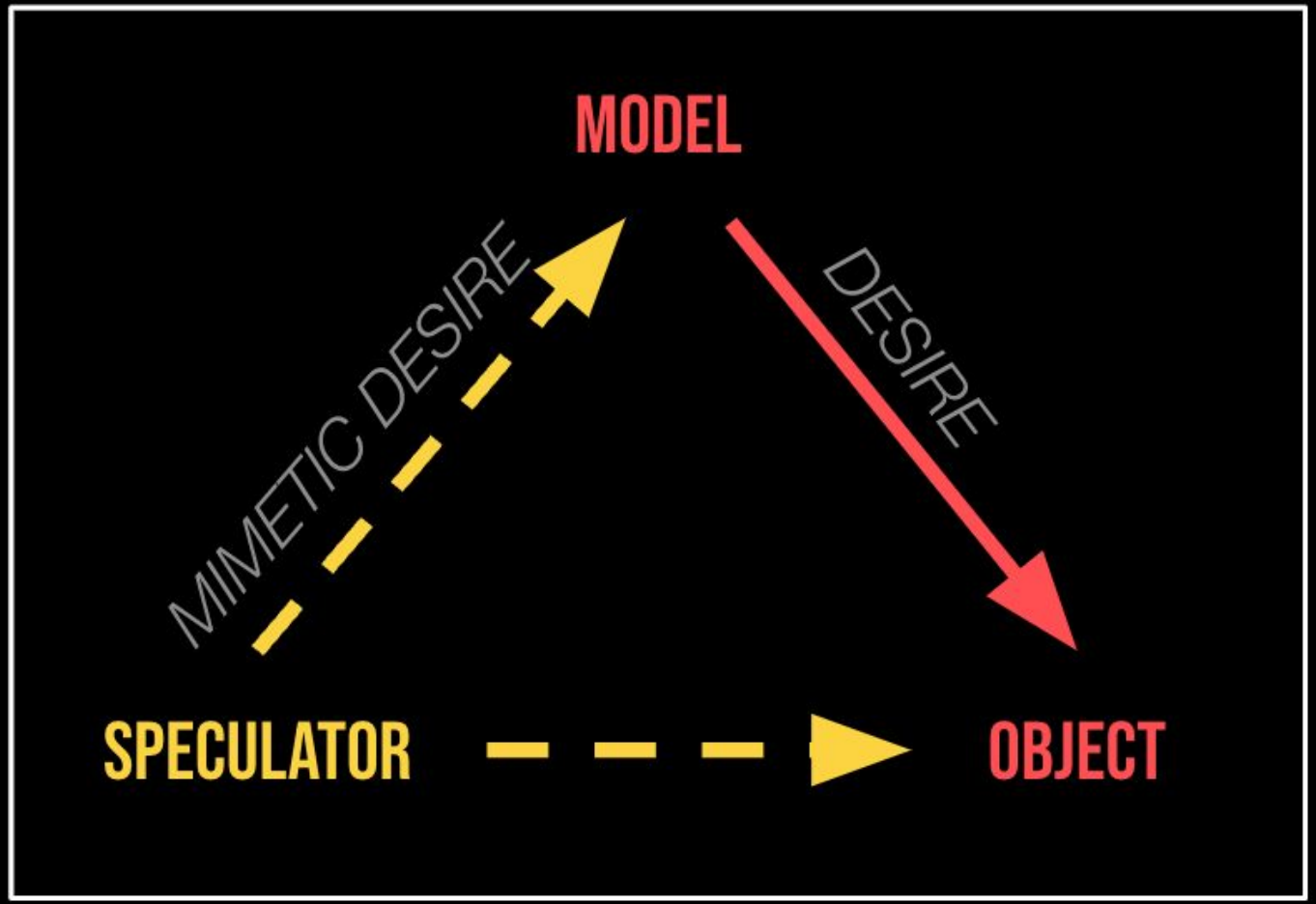
MIMETIC THEORY

The Bitcoin Handbook



Our desires are not intrinsic, but derived from the desires of others that we seek to imitate

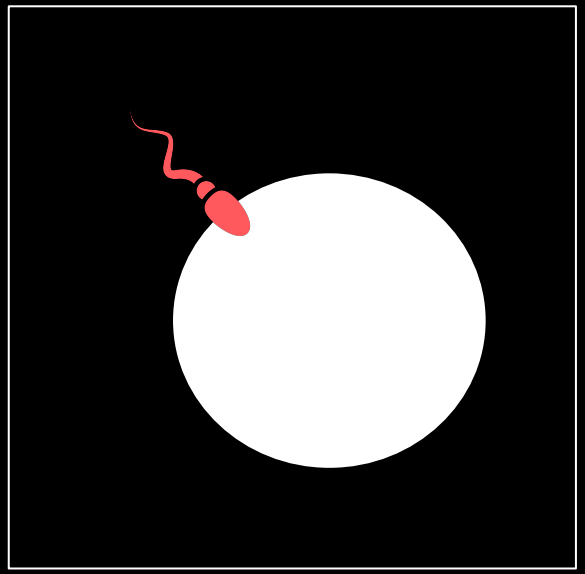
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FIRST CONCLUSION BIAS

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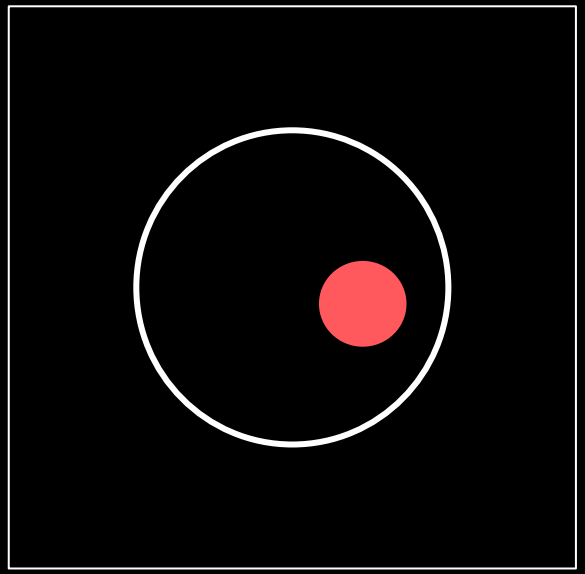
The tendency to accept the first answer offered, regardless of correctness, shutting out any further inquiry or debate

“The mind works a bit like a sperm and egg: the first idea gets in and then the mind shuts… leads us to accept many erroneous results.”

—CHARLIE MUNGER

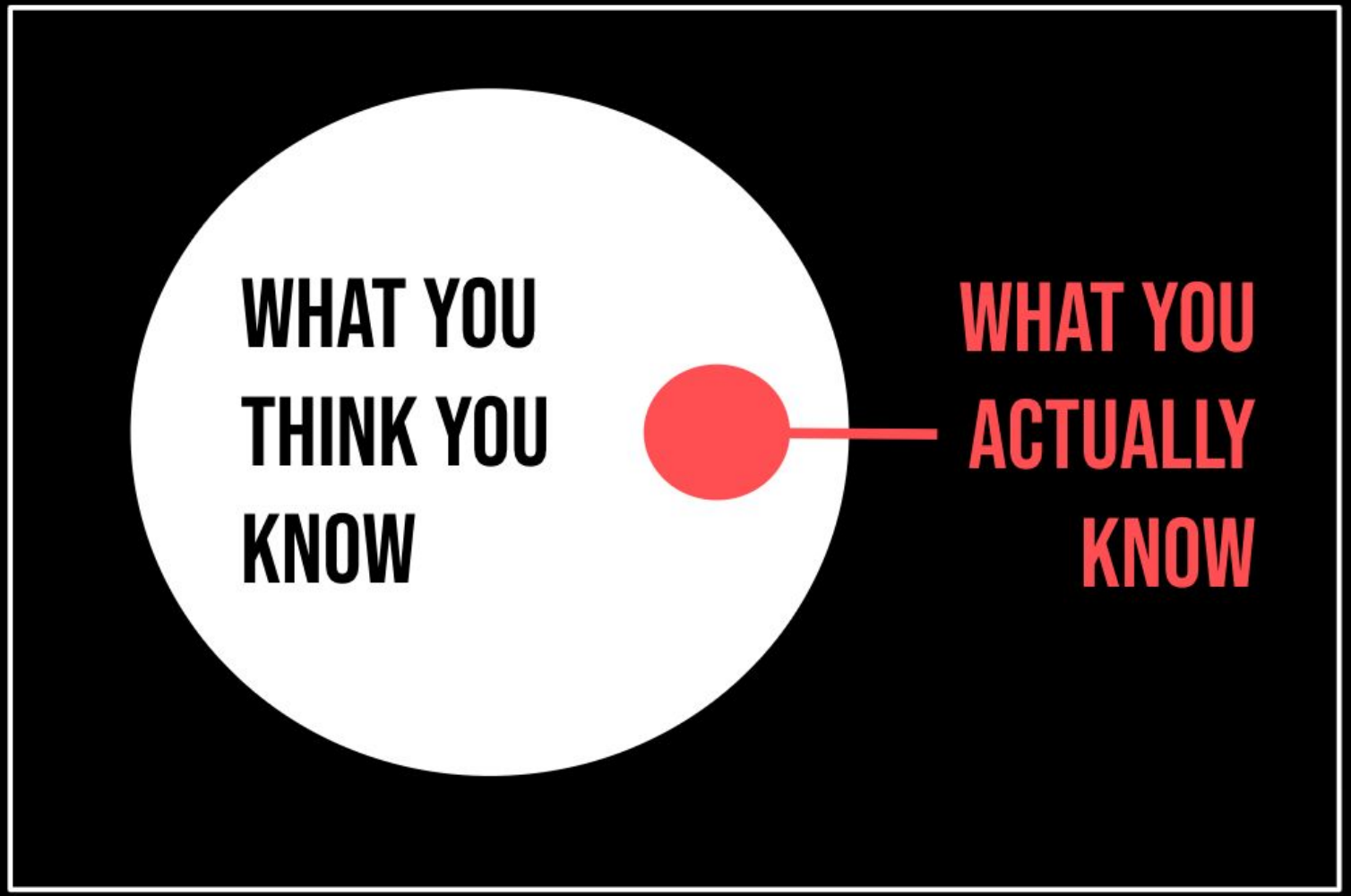
CIRCLE OF COMPETENCE

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Understanding where your knowledge is lacking will illuminate where you're vulnerable

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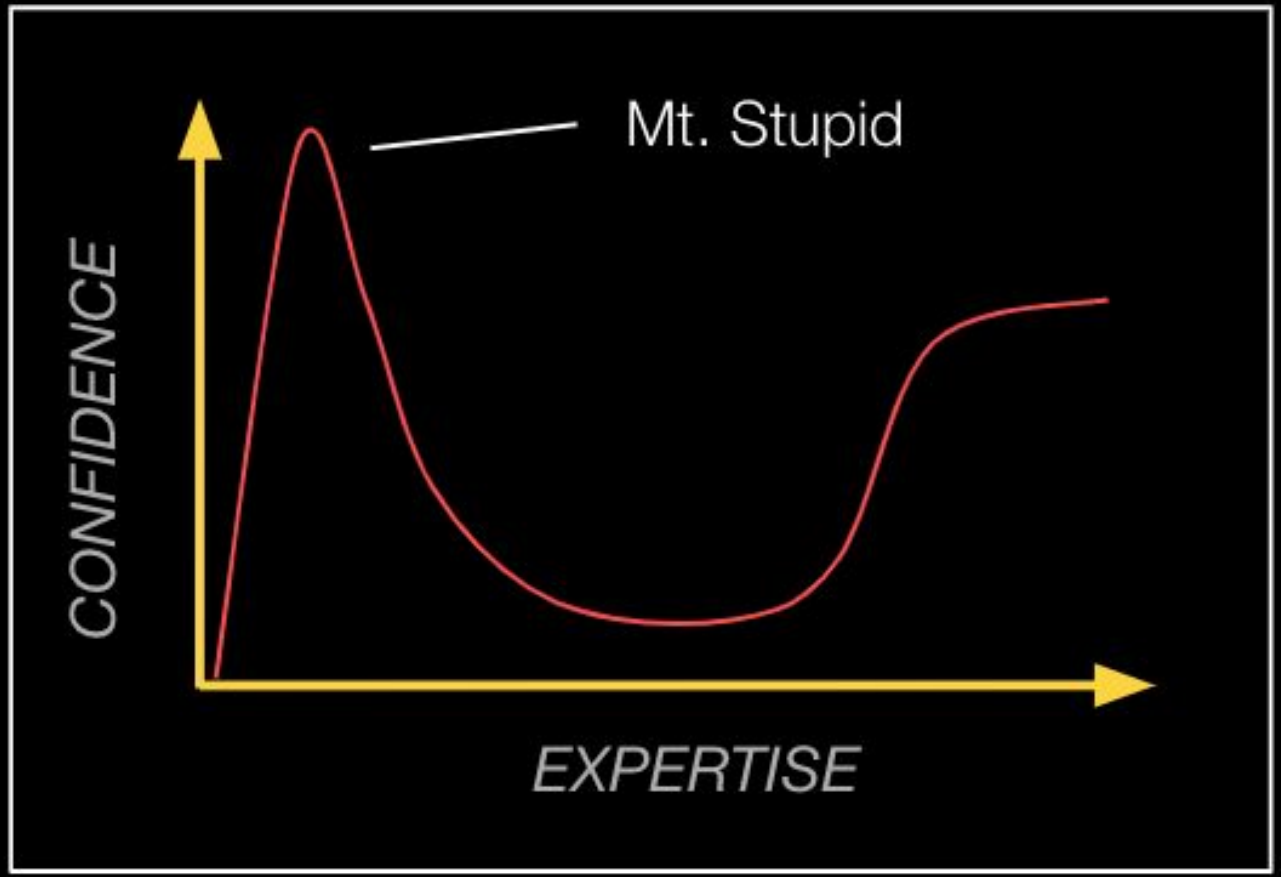
DUNNING-KRUGER EFFECT

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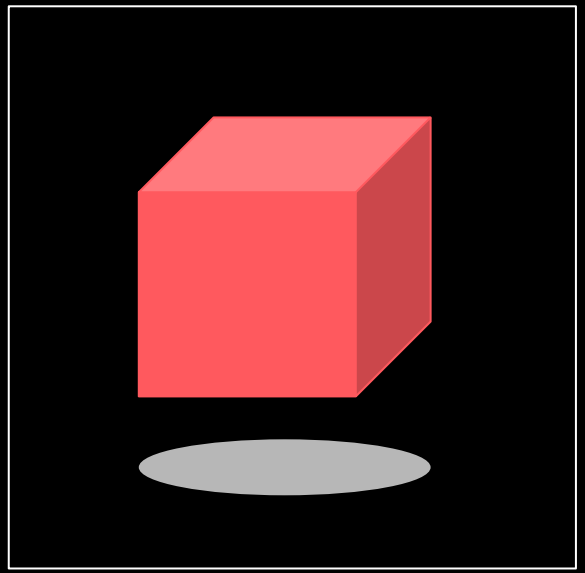
Overestimating your knowledge in a specific area leads to erroneous decisions



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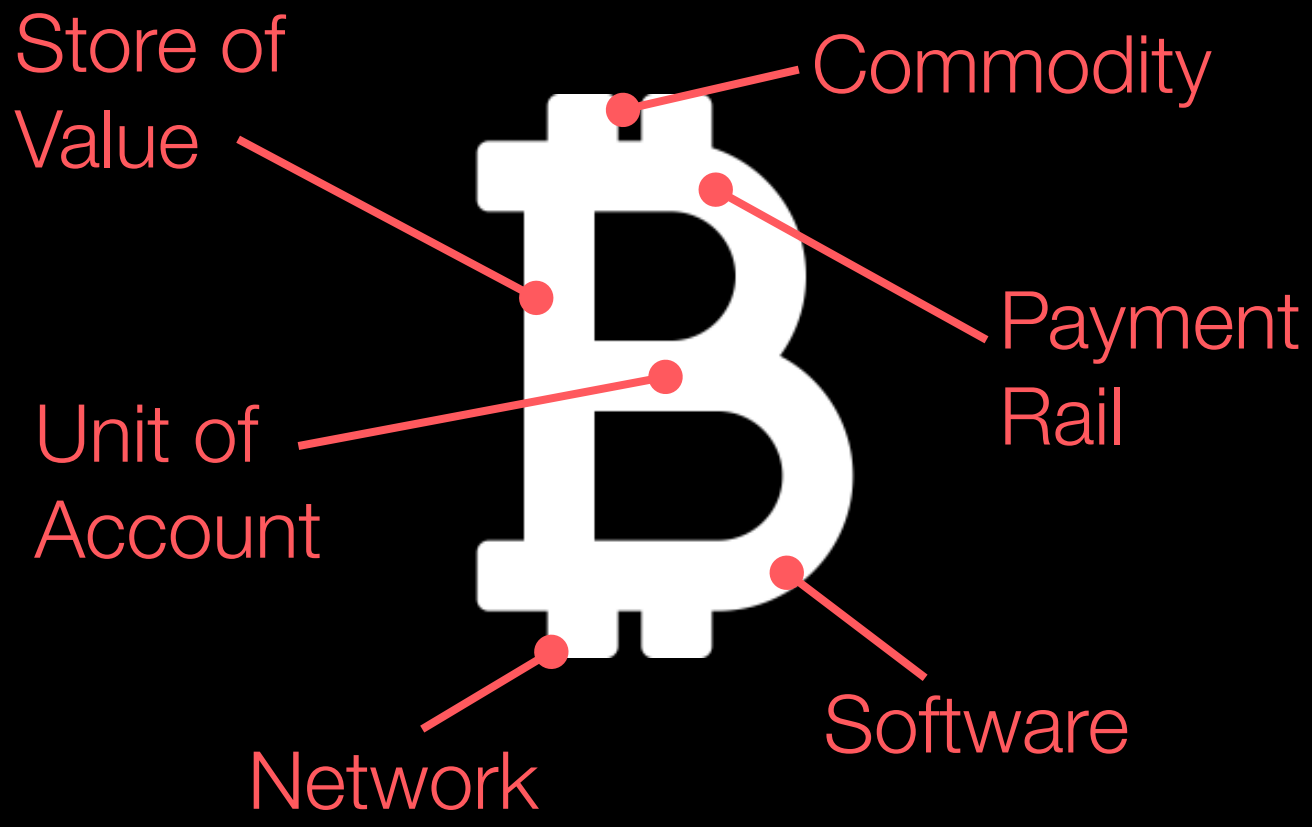
CATEGORY ERROR

The Bitcoin Handbook



Assuming an individual property of something is representative of its whole, leading to misattribution

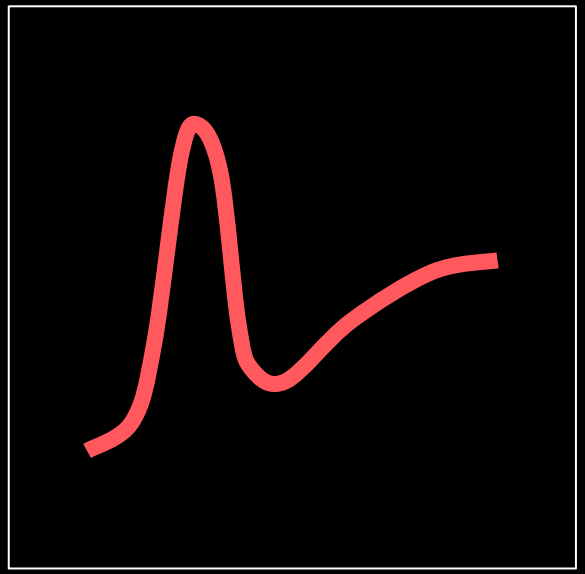
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Store of Value
Commodity
Payment Rail
Unit of Account
Network
Software

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GARTNER HYPE CYCLE

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A methodology for ascertaining the maturity and adoption of new technologies and applications

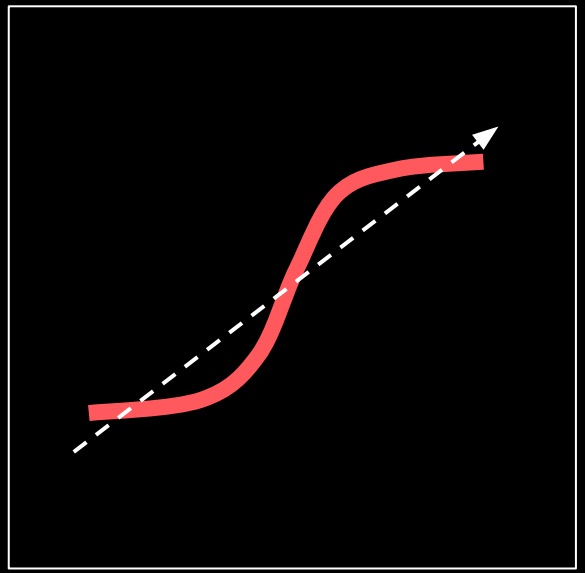
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EXPECTATIONS
Innovation Trigger
Peak of Inflated Expectations
Trough of Disillusionment
Slope of Enlightenment
Plateau of Productivity
TIME

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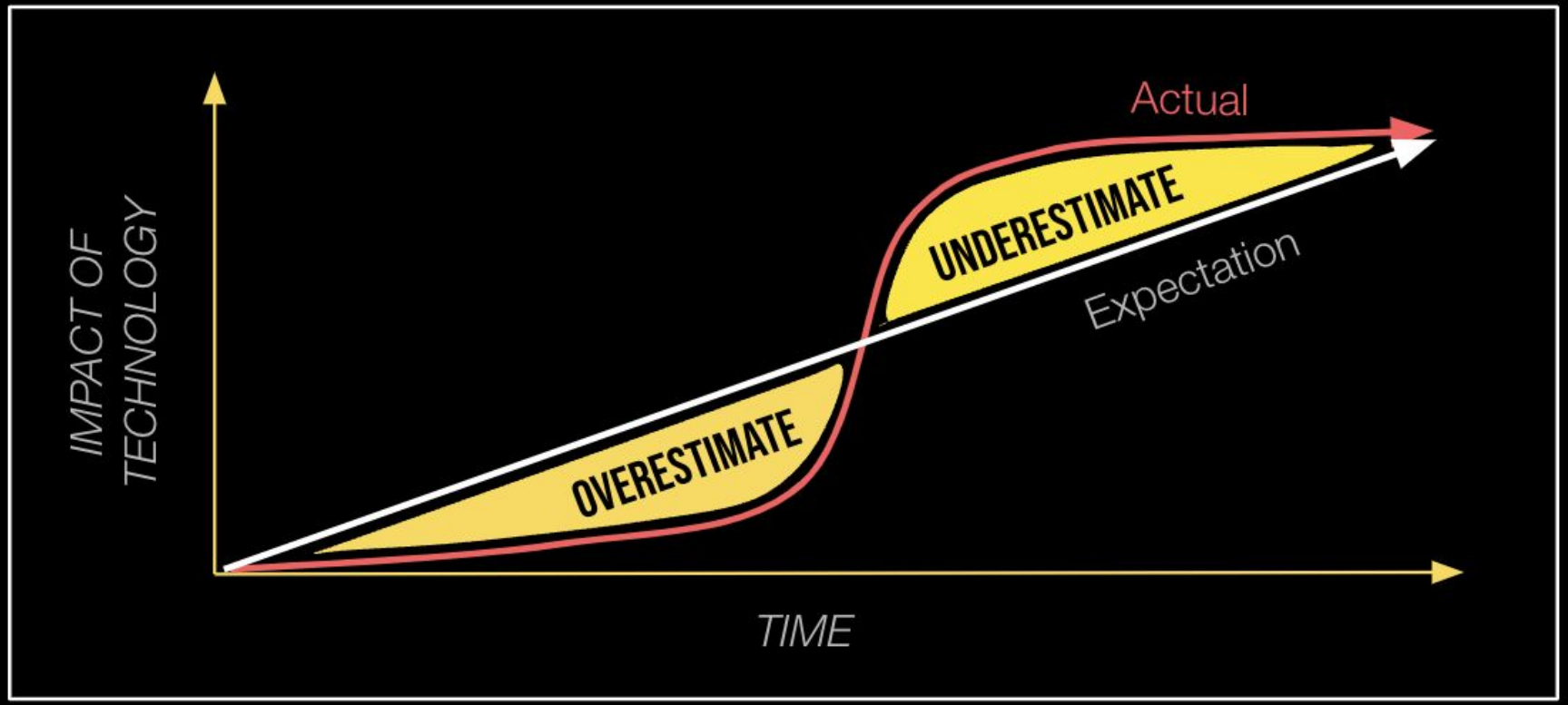
AMARA'S LAW

BITCOIN
ESSENTIAL CONCEPTS



We tend to overestimate the impact of new technologies in the short term and underestimate it in the long term

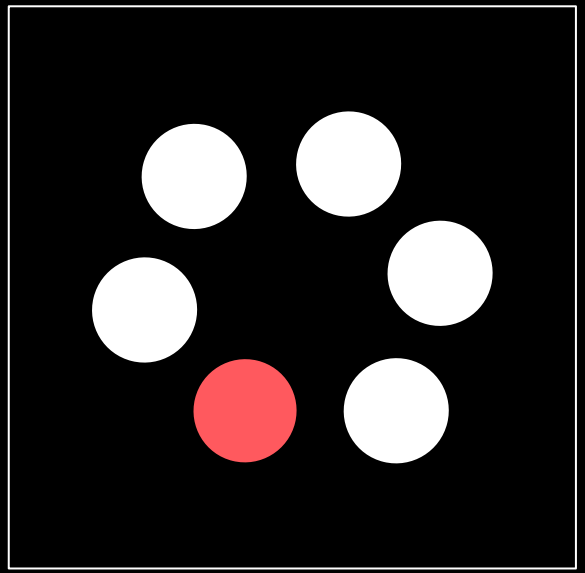
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IMPACT OF TECHNOLOGY
Actual
UNDERESTIMATE
Expectation
OVERESTIMATE
TIME

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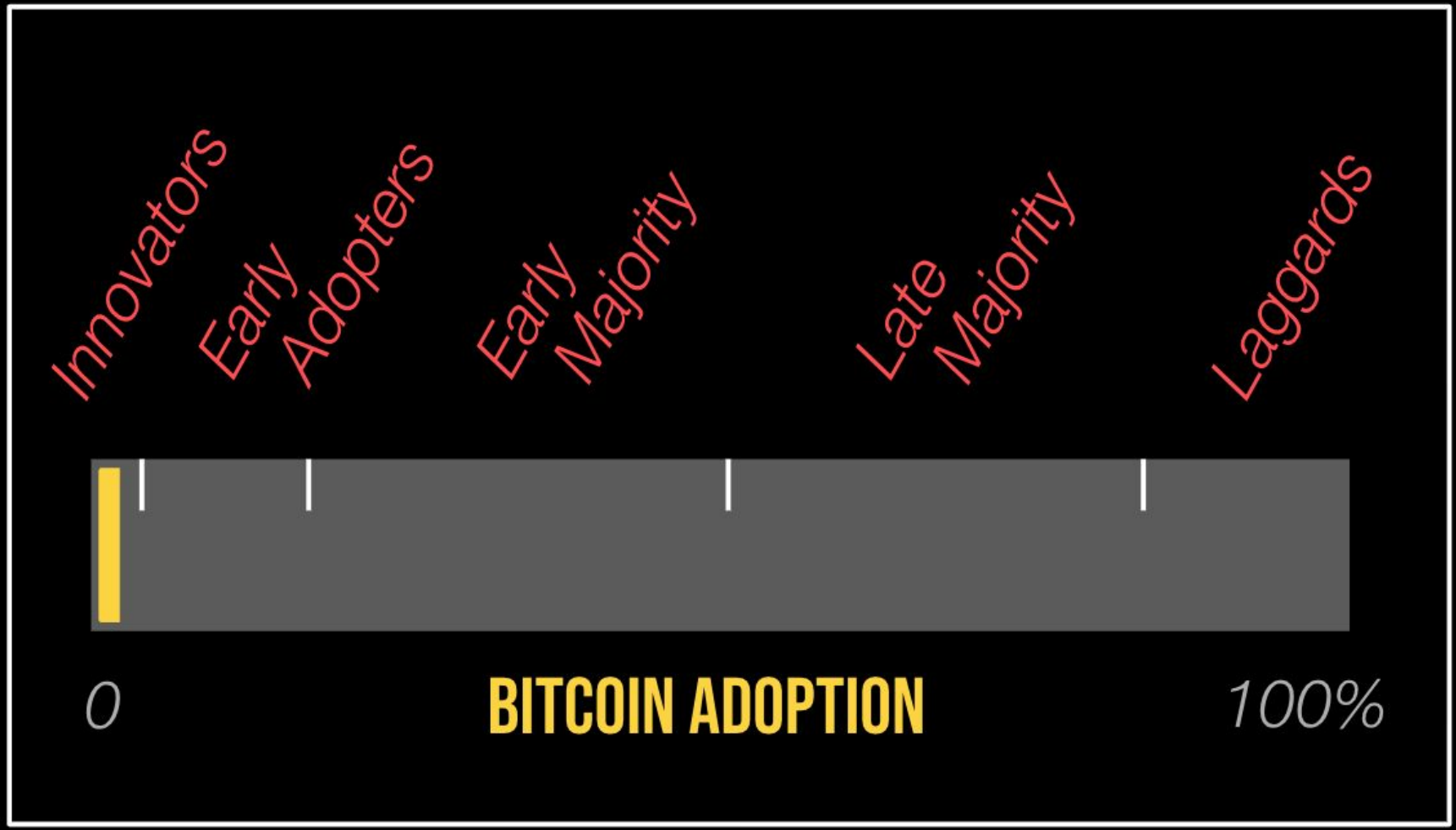
INTRANSIGENT MINORITY

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When a few individuals shape the preferences of the majority by their unwillingness to submit to the default option

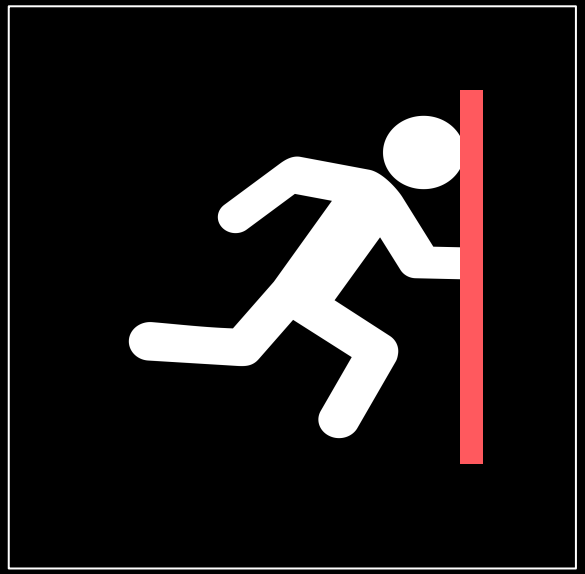
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Innovators
Early Adopters
Early Majority
Late Majority
Laggards
0
BITCOIN ADOPTION
100%

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INEQUIVALENCE THEOREM

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When economic conditions become excessively burdensome, individuals with the capacity will permanently exit a jurisdiction



"In the Information Age.. the rational person will not respond to the prospect of higher taxes to fund deficits..

Sovereign individuals and other rational people will flee jurisdictions with large unfunded liabilities."

–JAMES DALE DAVIDSON & WILLIAM REES-MOGG

GELL-MANN AMNESIA

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Observing errors in reporting of a topic, yet trusting the same source in areas where you are less knowledgeable

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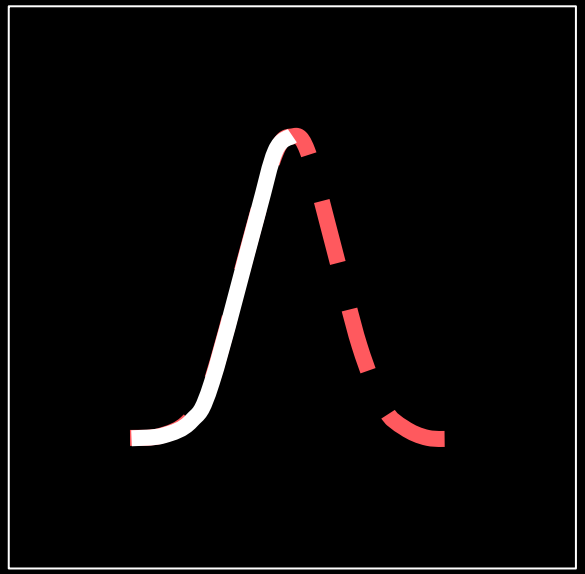
Newsweek

Bitcoin Mining on Track to Consume All of the World's Energy by 2020

ANTHONY CUTHBERTSON 12/11/17

THE LINDY EFFECT

The Bitcoin Handbook



The theory that the lifespan of non-perishable things are proportional to their current age

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3,468

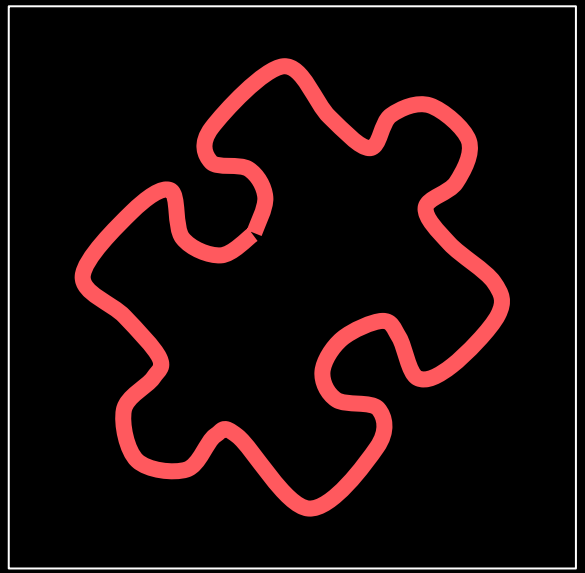
DAYS SINCE THE BITCOIN NETWORK HAS BEEN DOWN

As of 8 September 2022

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EMERGENCE

The Bitcoin Handbook

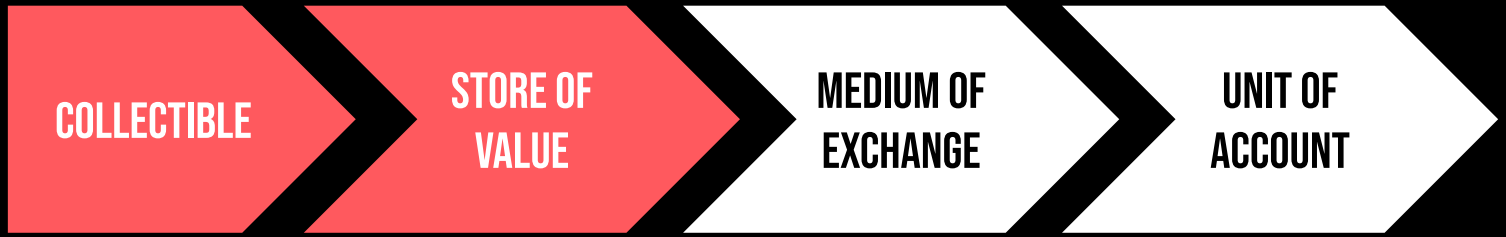


When novel dynamic structures arise from the interaction of individual components and becomes self-regulating

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“Bitcoin is currently transitioning from the first stage of monetization to the second stage.. No one alive has seen the real-time monetization of a good (as is taking place with Bitcoin), so there is precious little experience regarding the path this monetization will take.”

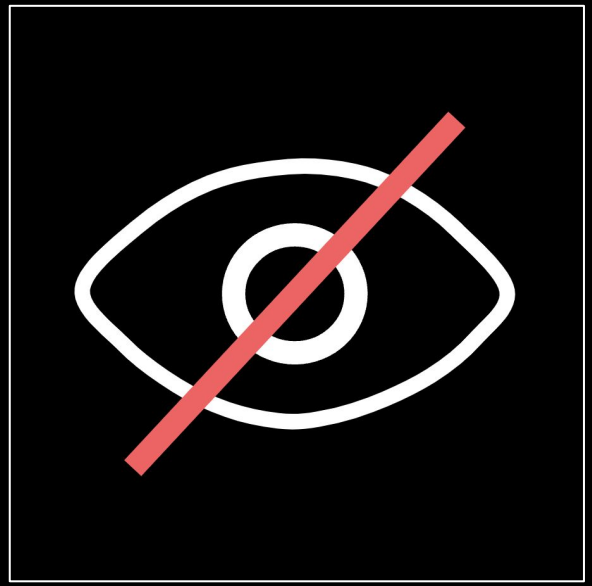
–VIJAY BOYAPATI



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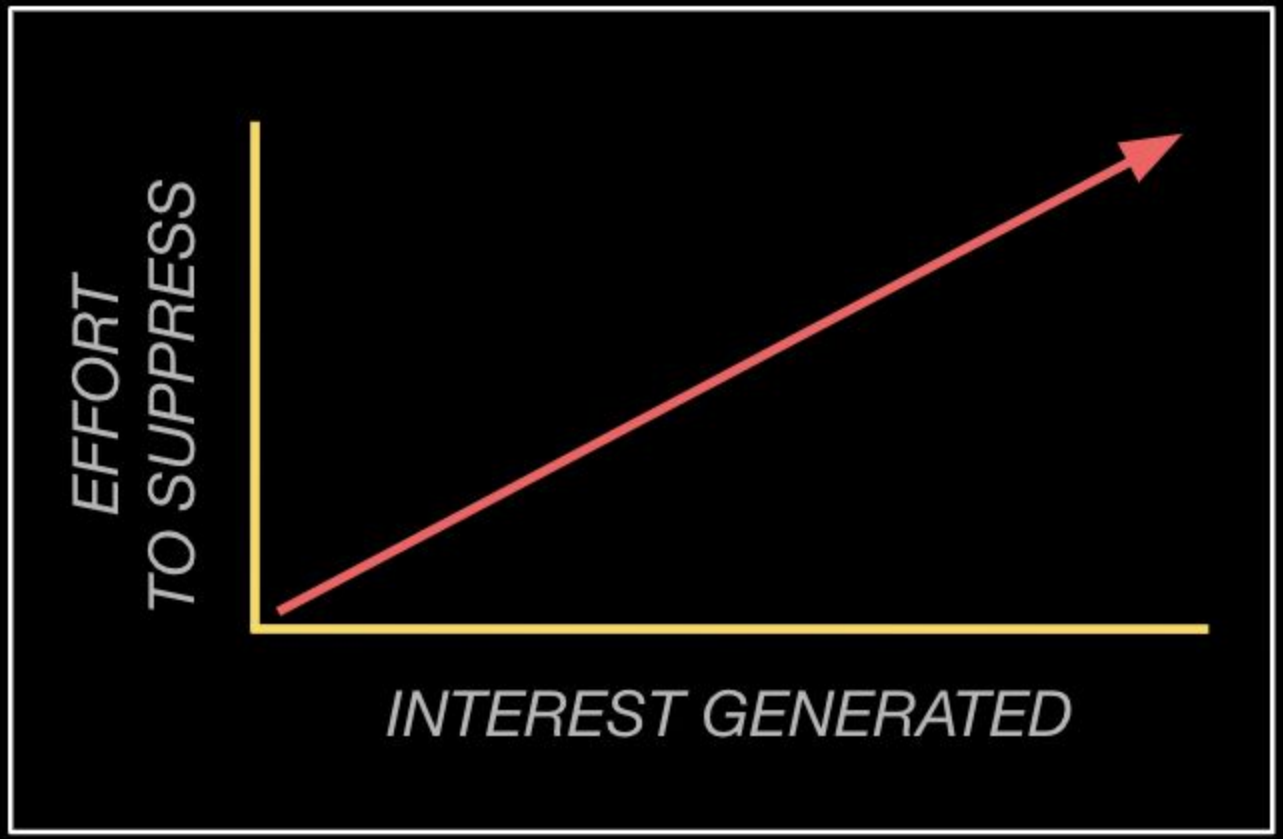
STREISAND EFFECT

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Attempting to hide, remove, or censor something has the unintended consequence of publicizing it more widely

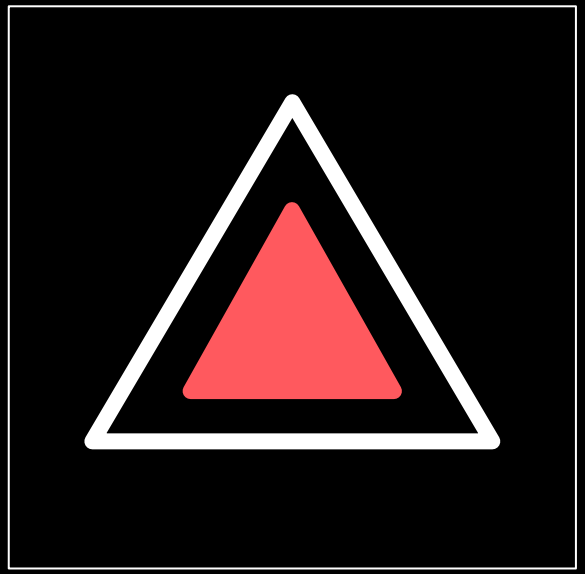
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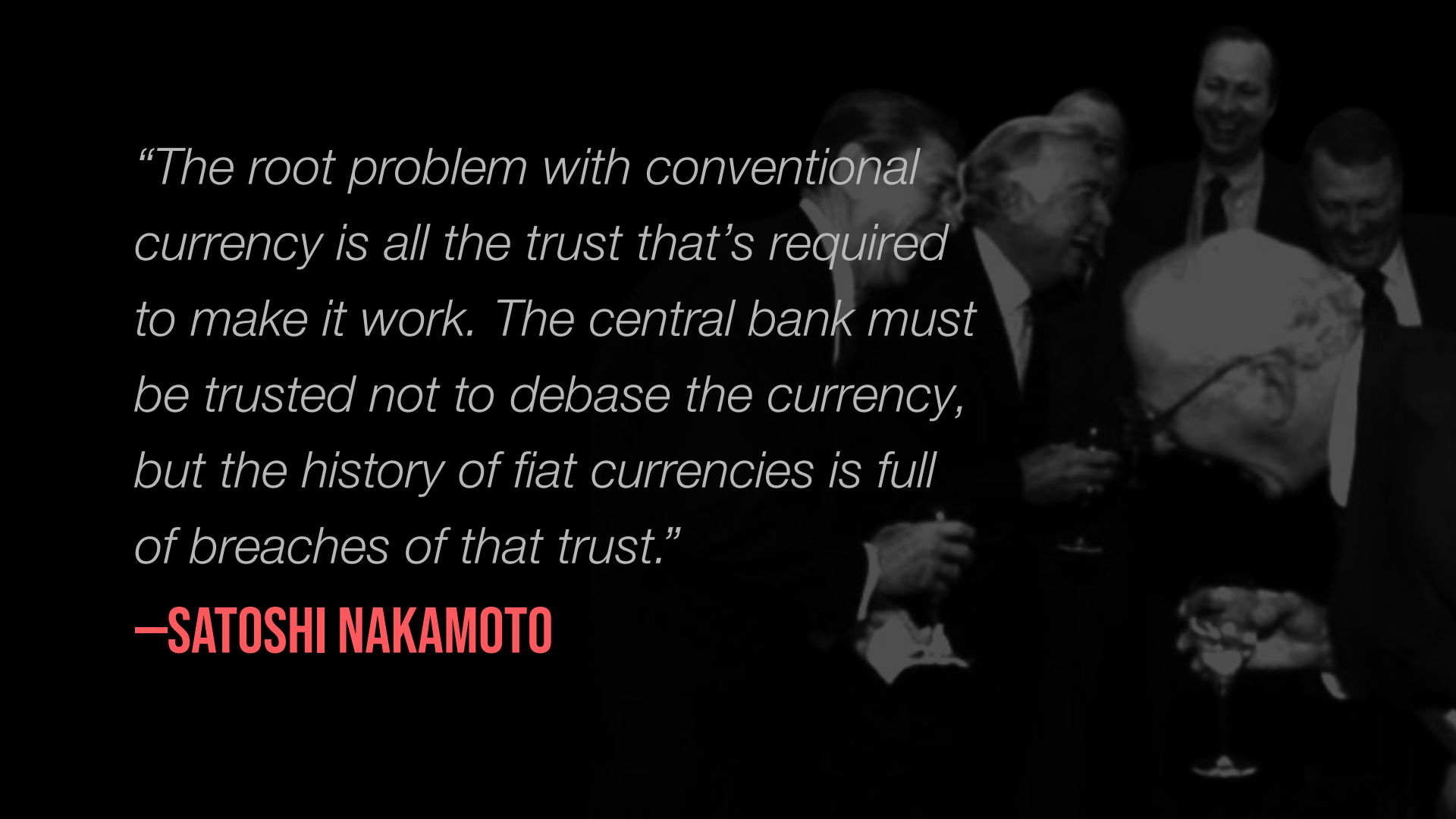
MORAL HAZARD

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When a party has a duty to serve the interests of others, but instead places its own interests first

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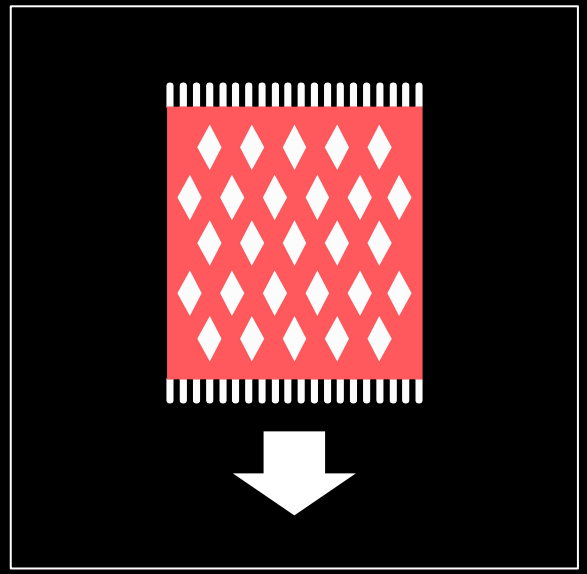


“The root problem with conventional currency is all the trust that’s required to make it work. The central bank must be trusted not to debase the currency, but the history of fiat currencies is full of breaches of that trust.”

—SATOSHI NAKAMOTO

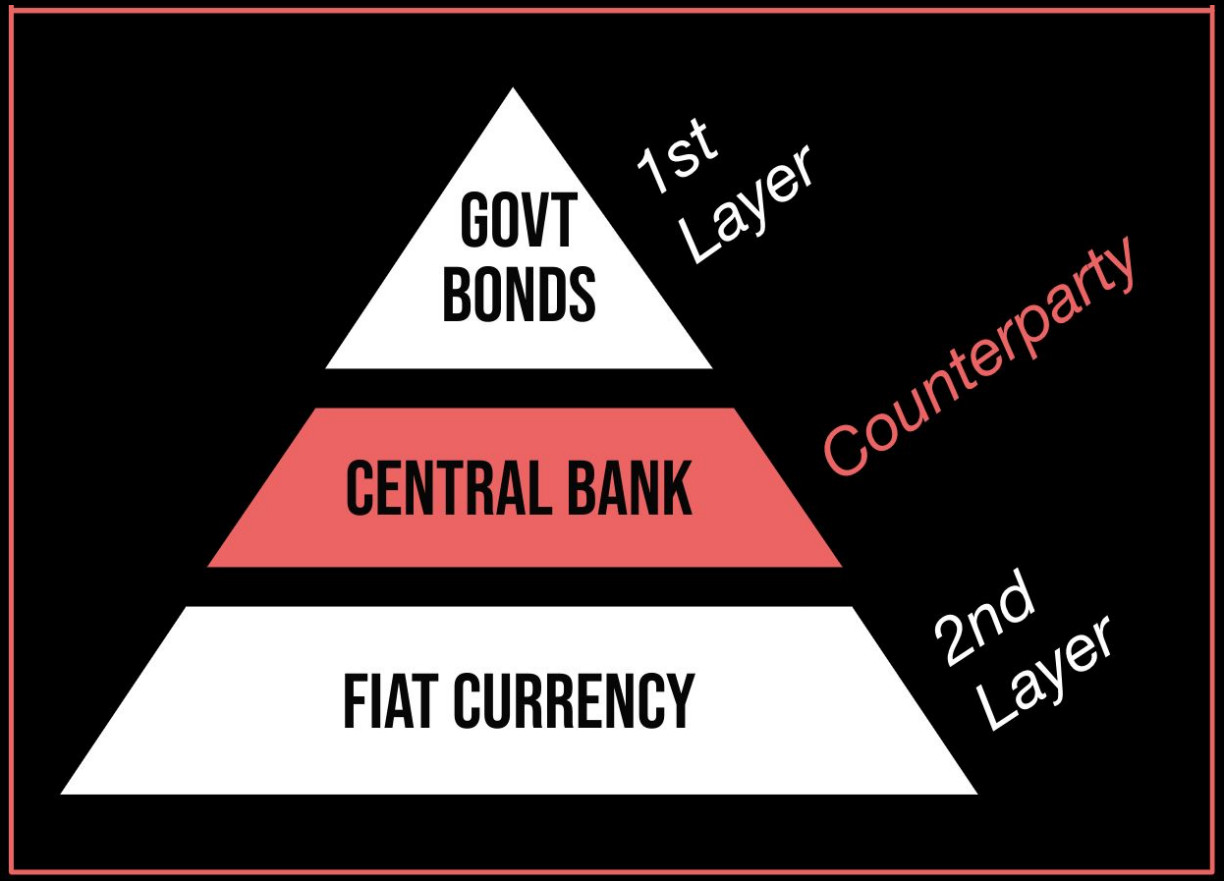
COUNTERPARTY RISK

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The likelihood of a party defaulting on its obligation in a transaction



Adapted from *Layered Money* by Nik Bhatia

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Anil Patel

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